

HCMC UNIVERSITY OF TECHNOLOGY AND ENGINEERING
FACULTY OF ELECTRICAL - ELECTRONICS ENGINEERING
DEPARTMENT OF COMPUTER AND COMMUNICATIONS
ENGINEERING



GRADUATION THESIS

DESIGN A NOISE-SUPPRESSED SWITCH BOX
FOR THE SEMICONDUCTOR TESTING
SOLUTION

COMPUTER ENGINEERING MAJOR

Student: **DANG TRUONG HUY**

ID: 21119202

HO CHI MINH CITY - 06/2026

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Instructor: **DAU TRONG HIEN, M.Sc**

HO CHI MINH CITY - 06/2026

**BỘ GIÁO DỤC VÀ ĐÀO TẠO
TRƯỜNG ĐẠI HỌC CNKT TP HCM
KHOA ĐIỆN-ĐIỆN TỬ
BỘ MÔN KT-MÁY TÍNH VIỄN THÔNG**

**CỘNG HÓA XÃ HỘI CHỦ NGHĨA VIỆT NAM
ĐỘC LẬP – TỰ DO – HẠNH PHÚC**

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Khóa: 2021

Giảng viên hướng dẫn: ThS. Đậu Trọng Hiền

Ngày nhận đề tài: 15/02/2026 **Ngày nộp đề tài:** 26/06/2026

1. Tên đề tài: **Design a noise-suppressed switch box for the semiconductor testing solution.**

Nội dung thực hiện đề tài:

- Thiết kế và chế tạo được bộ chuyển mạch giảm nhiễu để giải quyết vấn đề nhiễu loạn kết quả đo đặc trong kiểm thử bán dẫn
- Nghiên cứu cách thức lựa chọn kiến trúc chuyển mạch phù hợp.
- Nghiên cứu thiết kế sơ đồ nguyên lý, sơ đồ khối, sơ đồ vẽ mạch in PCB cho bộ chuyển mạch. Nghiên cứu lập trình mã nguồn nhúng vào vi điều khiển.
- Thực nghiệm sản phẩm thu thập kết quả đo đặc. Từ đó phân tích và đánh giá kết quả thu thập đó nhằm chứng minh bộ chuyển mạch có hiệu quả trong việc giảm nhiễu trong đo đặc.
- Đưa ra kết luận, ưu nhược điểm của đề tài, hướng phát triển của đề tài.

GIẢNG VIÊN HƯỚNG DẪN

BỘ MÔN KT MÁY TÍNH VIỄN THÔNG

BẢN NHẬN XÉT KHÓA LUẬN TỐT NGHIỆP
(Dành cho giảng viên hướng dẫn)

Đề tài: Design a noise-suppressed switch box for the semiconductor testing solution

Sinh viên: + Đặng Trường Huy

MSSV: 21119202

+

MSSV:

Hướng dẫn: Thạc sĩ. Đặng Trọng Hiền

Nhận xét bao gồm các nội dung sau đây:

1. Tính hợp lý trong cách đặt vấn đề và giải quyết vấn đề; ý nghĩa khoa học và thực tiễn:
Đặt vấn đề rõ ràng, mục tiêu cụ thể; đề tài có tính mới, cấp thiết; đề tài có khả năng ứng dụng, tính sáng tạo.

Có tính thực tiễn cao

2. Phương pháp thực hiện/ phân tích/ thiết kế:

Phương pháp hợp lý và tin cậy dựa trên cơ sở lý thuyết; có phân tích và đánh giá phù hợp; có tính mới và tính sáng tạo.

phù hợp

3. Kết quả thực hiện/ phân tích và đánh giá kết quả/ kiểm định thiết kế:

Phù hợp với mục tiêu đề tài; phân tích và đánh giá/ kiểm thử thiết kế hợp lý; có tính sáng tạo/ kiểm định chặt chẽ và đảm bảo độ tin cậy.

4. Kết luận và đề xuất:

Kết luận phù hợp với cách đặt vấn đề, đề xuất mang tính cải tiến và thực tiễn; kết luận có đóng góp mới mẻ, đề xuất sáng tạo và thuyết phục.

5. Hình thức trình bày và bố cục báo cáo:

Văn phong nhất quán, bố cục hợp lý, cấu trúc rõ ràng, đúng định dạng mẫu; có tính hấp dẫn, thể hiện năng lực tốt, văn bản trau chuốt.

6. Kỹ năng chuyên nghiệp và tính sáng tạo:

Thể hiện các kỹ năng giao tiếp, kỹ năng làm việc nhóm, và các kỹ năng chuyên nghiệp khác trong việc thực hiện đề tài.

7. Tài liệu trích dẫn

Tính trung thực trong việc trích dẫn tài liệu tham khảo; tính phù hợp của các tài liệu trích dẫn; trích dẫn theo đúng chỉ dẫn APA.

8. Đánh giá về sự trùng lặp của đề tài

Cán khẳng định đề tài có trùng lặp hay không? Nếu có, đề nghị ghi rõ mức độ, tên đề tài, nơi công bố, năm công bố của đề tài đã công bố.

9. Những nhược điểm và thiếu sót, những điểm cần được bổ sung và chỉnh sửa*

10. Nhận xét tinh thần, thái độ học tập, nghiên cứu của sinh viên

Đề nghị của giảng viên hướng dẫn

Ghi rõ: "Bảo cáo đạt/ không đạt yêu cầu của một khóa luận tốt nghiệp kỹ sư, và được phép/ không được phép bảo vệ khóa luận tốt nghiệp"

Được phép bảo vệ

Tp. HCM, ngày ... tháng ... năm 20...

Người nhận xét
(Ký và ghi rõ họ tên)

Đặng Trọng Hiền

* Giảng viên hướng dẫn có thể ghi tiếp vào mặt sau

BẢN NHẬN XÉT KHÓA LUẬN TỐT NGHIỆP

(Dùng cho giảng viên phản biện)

Đề tài:	Design a noise-suppressed switch box for the semiconductor testing solution		
Sinh viên:	Đặng Trường Huy	MSSV:	21119202

Nhận xét bao gồm các nội dung sau đây:

1. Tính hợp lý trong cách đặt vấn đề và giải quyết vấn đề; ý nghĩa khoa học và tính ứng dụng [15]:

Đặt vấn đề rõ ràng, mục tiêu cụ thể [5]; đề tài có tính mới, cấp thiết [5]; đề tài có khả năng ứng dụng, tính sáng tạo[5].

2.Phương pháp thực hiện/ phân tích/ thiết kế [25]:

Phương pháp hợp lý và tin cậy dựa trên cơ sở lý thuyết [10]; có phân tích và đánh giá phù hợp [10]; có tính mới và tính sáng tạo [5].

3. Kết quả thực hiện/ phân tích và đánh giá kết quả/ kiểm định thiết kế [25]:

Phù hợp với mục tiêu [10]; phân tích và đánh giá / kiểm thử thiết kế hợp lý [10]; có tính sáng tạo/ kiểm định chặt chẽ và tin cậy [5].

4. Kết luận và đề xuất [10]:

Kết luận phù hợp với cách đặt vấn đề, đề xuất cải tiến và thực tiễn [5]; kết luận có đóng góp mới mẻ, đề xuất sáng tạo và thuyết phục[5].

5. Hình thức trình bày, bố cục và chất lượng báo cáo [20]:

Văn phong nhất quán, bố cục hợp lý, cấu trúc rõ ràng, đúng định dạng mẫu [5]; có tính hấp dẫn, thể hiện năng lực tốt, văn bản trau chuốt [15].

6. Tài liệu trích dẫn [5]:

Tính trung thực trong việc trích dẫn tài liệu tham khảo; tính phù hợp của các tài liệu trích dẫn; trích dẫn theo đúng chỉ dẫn APA.

7. Đánh giá về sự trùng lặp của đề tài:

Cần khẳng định đề tài có trùng lặp hay không? Nếu có, ghi rõ mức độ, tên đề tài, nơi công bố, năm công bố của đề tài đã công bố.

8. Những nhược điểm và thiếu sót, những điểm cần được bổ sung và chỉnh sửa*:

Câu hỏi sinh viên phải trả lời trước hội đồng*:

Kết luận:	Cho bảo vệ	☐	Không cho bảo vệ*	☐	Điểm tổng:	/100
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*Trường hợp giảng viên phản biện không cho bảo vệ, vui lòng ghi rõ lý do:

TP. HCM, ngày tháng năm 2026

Người phản biện

ACKNOWLEDGEMENTS

First and foremost, I would like to express my thankful respects to these following individuals and organizations who have provided invaluable support during my graduation thesis:

- M.Sc. Đậu Trọng Hiền, my dedicated instructor, for his guidance and insightful idea throughout the research process. The knowledge and experience he shared with me have become a solid foundation, enabling me to apply them practically in the successful execution of this thesis.
- My family, for their unwavering support, encouragement, and provision of necessary resources during this thesis journey. Their unwavering support and unconditional love have been the greatest source of motivation, helping me recover through the most challenging times.
- All the teachers, lecturers from the Faculty of Electrical and Electronics Engineering for imparting fundamental and specialized knowledge during my years of study. These lessons have been essential in building the foundation for this project.

Throughout the process of conducting this thesis, I have always tried our best to work seriously. Although the project has been completed, there may still be errors. I truly hope to receive constructive feedback and comments from lecturers and fellow students so that I may improve this work further.

OVERVIEW

This graduation thesis focuses on the research and development of the noise-suppressed switch box design as the semiconductor testing solution. The primary objective of this project is to provide a solid understanding of semiconductor testing, to implement the practical hardware design together with the firmware optimization in the design. The project involved extensive research into the operational theory and parasitic effects optimization strategies for the switch box board design. Moreover, applying the Probability and Statistics method, the project evaluates and analyzes the test data samples which are collected during the real experimental testings, to prove that the switch box is able to suppress the parasitic effects.

Currently, the noise-suppressed switch box is successfully implemented, which is capable of controlling the switch relay signal between the test instrument HP 4280A and the test fixture based on the specific conditions. The main purpose of the switch box is to suppress the parasitic effects incurring to measurement at the sensitive level. The results are evaluated and calculated logically, thanks to a robust hardware foundation and an optimized testing methodology. This meets the initial objectives of cost reduction, increased accessibility, and suitable for educational and research purposes in the semiconductor testing.

TABLE OF CONTENTS

ACKNOWLEDGEMENTS	i
OVERVIEW	ii
LIST OF FIGURES.....	iii
LIST OF TABLES AND EQUATIONS	iviii
LIST OF ABBREVIATIONS	ix
CHAPTER 1. ABSTRACT.....	1
1.1. PROBLEM STATEMENT	1
1.2. RESEARCH OBJECTIVES.....	2
1.3. RESEARCH METHODS.....	2
1.4. RESEARCH LIMITS.....	3
1.5. RESEARCH CONTENTS	3
1.6. REPORT STRUCTURE	4
CHAPTER 2. THEORETICAL BASIS.....	5
2.1. OVERVIEW OF THE SEMICONDUCTOR TESTING.....	5
2.1.1. Overview of the semiconductor testing stage	5
2.1.2. The semiconductor testing process	6
2.1.3. The existing issue	13
2.1.4. The approaching method	16
2.2. COMMON SWITCH ARCHITECTURES	17
2.2.1. Overview of the switch architecture.....	17
2.2.2. No switching.....	18
2.2.3. Switching in Test Rack only	18
2.2.4. Switching in Test Fixture only	19
2.2.5. Switching in Test Rack and Test Fixture	20

2.3. COMMON SWITCH TOPOLOGIES	22
2.3.1. Overview of the switch topology	22
2.3.2. General-Purpose Relays	23
2.3.3. Multiplexers.....	24
2.3.4. Matrix	25
CHAPTER 3. SYSTEM DESIGN	27
3.1. SYSTEM DESIGN.....	27
3.1.1. System requirements	27
3.1.2. Block diagram	28
3.2. HARDWARE DESIGN	29
3.2.1. Requirements for hardware design.....	29
3.2.2. Schematic diagram	30
3.2.3. Layout PCB	38
3.3. SOFTWARE IMPLEMENTATION.....	42
3.3.1. Requirements for software implementation	42
3.3.2. Flowchart.....	43
CHAPTER 4. RESULTS	45
4.1. THE OUTCOME OF THE PROJECT.....	45
4.2. DEMO	50
4.3. DATA ANALYSIS	55
CHAPTER 5. CONCLUSION AND FUTURE DEVELOPMENT	63
5.1. CONCLUSION	63
5.2. PROS AND CONS.....	63
5.3. FUTURE DEVELOPMENT	64
REFERENCES.....	65

LIST OF FIGURES

Fig 2. 1. The global semiconductor packaging and test market growth (semiconductorinsight.com)	6
Fig 2. 2. The probe test reference (source: datatec.com)	7
Fig 2. 3. The package test reference (source: tek.com)	8
Fig 2. 4. ISO Certified Cleanroom (isocleanroomchina.com).....	8
Fig 2. 5. Process in various classes level of cleanroom	9
Fig 2. 6. The HP 4280A 1MHz C Meter / C-V Plotter test instrument	10
Fig 2. 7. The high-frequency test fixture reference.....	12
Fig 2. 8. Parasitic capacitance in PCB layout (source: vse.com)	13
Fig 2. 9. Crosstalk in PCB design (protoexpress.com).....	14
Fig 2. 10. Propagation delay in the switching signal (protoexpress.com)	16
Fig 2. 11. No Switching architecture (source: NI.com)	18
Fig 2. 12. Switching in Test Rack only (source: NI.com)	19
Fig 2. 13. Switching relays inside the test fixture only (source: NI.com)	20
Fig 2. 14. Hybrid Switching (source: NI.com).....	21
Fig 2. 15. Single-pole single-throw relay – SPST relay (source: NI.com)	23
Fig 2. 16. Single-pole double-throw relay – SPDT relay (source: NI.com)	24
Fig 2. 17. Double-pole single-throw relay – DPST relay (source: NI.com)	24
Fig 2. 18. The 4 x 1 MUX of parallel level SPST relays (source: NI.com)	24
Fig 2. 19. The 4 x 1 MUX cascade level SPDT relays (source: NI.com)	25
Fig 2. 20. The 4x5 matrix topology in switching system (source: NI.com)	26

<i>Fig 3. 1. BNC connector in the switching system</i>	<i>27</i>
<i>Fig 3. 2. Overall block diagram of the system</i>	<i>28</i>
<i>Fig 3. 3. Detailed block diagram of the switch box</i>	<i>29</i>
<i>Fig 3. 4. Hardware design requirements</i>	<i>30</i>
<i>Fig 3. 5. Schematic diagram of MCU circuit in the relay driver board</i>	<i>31</i>
<i>Fig 3. 6. MCU ATmega328P Arduino Nano</i>	<i>31</i>
<i>Fig 3. 7. Schematic diagram of relay driver board connection pins</i>	<i>32</i>
<i>Fig 3. 8. DB25 male and female connector.....</i>	<i>33</i>
<i>Fig 3. 9. Overall schematic diagram of 12 opto-controllers circuit</i>	<i>34</i>
<i>Fig 3. 10. Detailed schematic diagram of opto-controller circuit</i>	<i>34</i>
<i>Fig 3. 11. PC817C optocoupler (source: PC817C datasheet)</i>	<i>35</i>
<i>Fig 3. 12. PC817C pin diagram (source: PC817 datasheet).....</i>	<i>35</i>
<i>Fig 3. 13. Overall schematic diagram of 12 relays matrix circuit.....</i>	<i>36</i>
<i>Fig 3. 14. Schematic diagram of relay matrix board connection pins.....</i>	<i>36</i>
<i>Fig 3. 15. Detailed schematic diagram of the relay matrix circuit.....</i>	<i>36</i>
<i>Fig 3. 16. IM03CGR relay (source: TE Axicom)</i>	<i>37</i>
<i>Fig 3. 17. Pin diagram of the IM03CGR relay (source: IM03CGR datasheet)...</i>	<i>37</i>
<i>Fig 3. 18. BNC connector structure (metabee.com)</i>	<i>38</i>
<i>Fig 3. 19. Top layer PCB layout for the relay matrix board</i>	<i>39</i>
<i>Fig 3. 20. Bottom layer PCB layout for the relay matrix board</i>	<i>39</i>
<i>Fig 3. 21. Top layer PCB layout for the relay driver board</i>	<i>40</i>
<i>Fig 3. 22. Bottom layer PCB layout for the relay driver board.....</i>	<i>40</i>
<i>Fig 3. 23. Flowchart of the switching relay system</i>	<i>44</i>

Fig 4. 1. <i>Top layer 3D view of the driver relay board</i>	45
Fig 4. 2. <i>Bottom layer 3D view of the driver relay board</i>	45
Fig 4. 3. <i>Top layer 3D view of the relay matrix board</i>	46
Fig 4. 4. <i>Bottom layer 3D view of the relay matrix board</i>	46
Fig 4. 5. <i>The front side of the relay matrix board</i>	47
Fig 4. 6. <i>The behind side of the relay matrix board</i>	47
Fig 4. 7. <i>The front side of the relay driver board</i>	48
Fig 4. 8. <i>The behind side of the relay driver board</i>	48
Fig 4. 9. <i>The top side of the switch box</i>	49
Fig 4. 10. <i>The left side of the switch box</i>	49
Fig 4. 11. <i>The side position of the switch box</i>	50
Fig 4. 12. <i>Send demo commands to switch signal relays</i>	50
Fig 4. 13. <i>Two capacitor relays turn on after commands</i>	51
Fig 4. 14. <i>Setup connection for demo testing</i>	52
Fig 4. 15. <i>Send commands to turn on capacitor meter relays and port relays</i>	53
Fig 4. 16. <i>Data recorded in capacitance mode on the test instrument's screen..</i>	54
Fig 4. 17. <i>Line graph of 10 datas recorded while testing with the switch box</i>	55
Fig 4. 18. <i>Standard deviation</i>	56
Fig 4. 19. <i>Data measurement value while testing with no switch box solution ...</i>	58
Fig 4. 20. <i>Line graph of 10 datas recorded while testing without the switch box</i>	59
Fig 4. 21. <i>Comparison between two of line graphs</i>	61

LIST OF TABLES AND EQUATIONS

Table 2. 1. Typical specifications set before testing	10
Table 2. 2. The trade-off in the choice of four switching architectures	22
Table 3. 1. The MCU Arduino Nano pins connector	32
Table 3. 2. DB25 connector pins	33
Table 3. 3. Pin definition of MCU.....	42
Table 3. 4. The input serial string definition	43
Table 4. 1. List of items needed in the second demo phase	51
Equation 2. 1. <i>parallel plate capacitance formula</i>	13
Equation 2. 2. <i>Electric field energy formula</i>	14
Equation 2. 3. <i>Magnetic field energy formula</i>	15
Equation 2. 4. <i>Propagation delay formula</i>	15
Equation 3. 1. <i>Led resistane Formula</i>	35
Equation 4. 1. <i>The Mean formula</i>	55
Equation 4. 2. <i>Standard Deviation formula</i>	57

LIST OF ABBREVIATIONS

No.	Abbre.	Meaning
1.	AC	Alternating Current
2.	BNC	Bayonet Neill-Concelman connector
3.	DC	Direct Current
4.	DUT	Device Under Test
5.	DPST	Double-Pole Single-Throw
6.	DPDT	Double-Pole Double-Throw
7.	EDS	Electrical Die Sorting
8.	ENIG	Electroless Nickel and Immersion Gold
9.	ESR	Equivalent Series Resistance
10.	GPIO	General Purpose Input Output
11.	HEPA	High Efficiency Particulate Air filter
12.	HI	High Input
13.	ICs	Integrated Circuits
14.	ISO	International Organization of Standardization
15.	IoT	Internet Of Things
16.	MCU	Micro Controller Unit
17.	MUX	Multiplexer
18.	OSAT	Outsourced Semiconductor Assembly and Test
19.	PCB	Printed Circuit Board
20.	PPE	Personal Protective Equipment
21.	SOT-23	Small Outline Transistor - 23
22.	SPST	Single-Pole Single-Throw
23.	SPDT	Single-Pole Double-Throw
24.	SRAM	Static Random Access Memory
25.	SMT	Surface Mount Technology
26.	RH	Relative Humidity

CHAPTER 1. ABSTRACT

1.1. PROBLEM STATEMENT

The semiconductor testing is a non-trivial process of the overall semiconductor industry. At its core, semiconductor testing is a process of making precise measurements of electrical characteristics of a device under test under specific conditions. This back-end stage inspects whether devices under test operate properly or not. Measurement is the foundation of semiconductor testing. But not all times, measurement process is completely smooth. At picoFarad-level capacitance measurement for devices under test, the value is extremely sensitive to noise. Causes of noise mainly comes from manual physical connections and the parasitic capacitance of test equipments. Everytime unplugging and replugging cables between the test instrument and the test fixture manually leads to fluctuations in value reading, unreliable production accuracy. Moreover, the unwanted parasitic capacitance from additional cables, signal lines can result in contact resistance noise, inefficient measurement.

To solve this pressing issue, the approaching solution is to implement a suppressed-noise switch controller that auto-facilitate this signal routing between the test instrument and test fixture while minimizing resistance noise as much as possible. For this reason, I chose the project “Design a noise-suppressed switching box for the semiconductor testing solution”. In terms of necessity, this project has strong practical value because it addresses real demands which optimizes the test process in my company. In terms of novelty, the similar product has previously been built by some individuals, but their work remained at the prototype stage without formal report, reference documentation. Therefore, this thesis currently improves limitations of earlier attempts through delivering a structured, academic study report.

1.2. RESEARCH OBJECTIVES

Four objectives of the project are presented as follows:

Firstly, implementing the switching PCB board to control relay switching signals between the test instrument and the test fixture. This is the main function of switch controller.

Secondly, the design needs to follow the suitable switching architecture, proper layout principles, strategic routing techniques and suitable component selection for noise optimization.

Thirdly, the project analyzes and compares experimental results, including parameter measurement and data processing. Ensure to prove the result to be scientific and reliable.

Fourthly, the project ensures that the product is easy to use, low-cost, well-suited size and optimized for practical application.

1.3. RESEARCH METHODS

To carry out the project effectively and logically, I applied several research methods as follows:

Firstly, document research method: This involves searching for reference from various reliable sources such as books, journals, the Internet, official manufacturer websites, datasheets, user manuals, scientific articles, etc., while also leveraging basic knowledge to study, learn, and conduct deeper analysis to solve the problem.

Secondly, experimental method: This involves testing the designed product, collecting and recording data and parameters under the specific conditions.

Thirdly, analytical and synthetic method: This method breaks down the problem into smaller components, analyzes the strengths, weaknesses, and attributes of each sub-problem. Through this, a clearer understanding is gained, which links the core principles and draw conclusions.

1.4. RESEARCH LIMITS

Due to the specific conditions for testing process, the project has the following limitations:

Firstly, design orientation: the orientation of project is focus on PCB hardware design integrated with firmware implementation, not focus on Internet of Things (IoT).

Secondly, scope limitation: The test fixture type using in this project is only testing for passive device under test, package diodes. Also, the test fixture has BNC connectors that suitable for HP 4280A test equipment terminals. The test instrument using in this project is HP 4280A test instrument.

Thirdly, environment requirements: Requirement for temperature and humidity is capped at 23 ± 1 °C, 40–60 % RH. If exceeds this specific conditions, the result can be changed. So, the temperature is accepted around 22 to 25 °C, the humidity is around 50 % RH, which is strictly controlled in the cleanroom.

Fourthly, testing instrument condition: The project uses the inner available DC bias source of the HP 4280A test instrument, not using an external DC source.

1.5. RESEARCH CONTENTS

Research content includes:

Firstly, overview study: A basic understanding of the semiconductor testing, the challenge and solution in the testing process. The position, role and the main purpose of the testing process in semiconductor industry, the switching architecture and common switching topologies.

Secondly, in-depth study: Based on datasheets, reference documents, and scientific papers, analyze the advantages and disadvantages of switching architectures; Consider the best choice of common switch topologies; Operating principle and role of general-purpose relays.

Thirdly, statistical study: Investigate statistical methods to collect test data to draw conclusion and make informed decisions.

- Hardware implementation: Design and explain schematic diagrams of each hardware component block of the switch box system; Draw PCB design into actual board.
- Software development: Develop algorithms, write embedded firmware code to hardware device.

1.6. REPORT STRUCTURE

Report structure has 5 chapters:

Chapter 1: Abstract. This chapter provides an overview of the research direction, including the reason for choosing the topic, research methods, research objectives, research content, project limitations, and the overall structure of the report.

Chapter 2: Theoretical basis. This chapter covers essential theoretical concepts related to the semiconductor testing process, challenge and solution; Overview the switch architecture and common switch topologies; Explain the advantages and disadvantages of each parts, from that explain the component selection for noise optimization in testing process.

Chapter 3: System design. This chapter details the system requirements and component selection. It includes the hardware design process, schematic diagrams, PCB layout, and the software design flow. Additionally, algorithm flowcharts are presented and explained to clarify the software's operational logic.

Chapter 4: Results. This chapter presents the outcomes achieved during the project. It includes the actual model results, experimental testing of the measurement system, recording measurement results and analyzing the recorded data.

Chapter 5: Conclusion and Future development. This final chapter draws conclusions based on the obtained results, discusses the strengths and limitations of the project, and proposes potential directions for future improvements and development.

CHAPTER 2. THEORETICAL BASIS

2.1. OVERVIEW OF THE SEMICONDUCTOR TESTING

2.1.1. Overview of the semiconductor testing stage

The semiconductor industry involves three primary stages: Integrated Circuits (ICs) Design, Fabrication and Packaging & Test. In the scope of the project, we only focus on the semiconductor testing stage. If the previous stage – Fabrication stage is called the Front-End stage, the Packaging & Test stage is called the Back-End stage. After the Front-End stage is complete, the silicon wafers are still not ready to be used for electronic devices. They must continue to go through the Back-End stage, which consists of Packaging and Final Test. The packaging stage refers to the process that helps the tiny chips covered with the protective package, also helps connect to the outside world. This stage usually includes some mini processes such as Wafer Dicing, Die Attach, Wire Bonding, Encapsulation, ... [1].

The test stage refers to the process that verifies the chips meet all the datasheet specifications before coming to the market. This stage usually includes some mini processes such as Wafer Probe Test, Package Test (Final Test), Burn-in Test, ... The company conducts this phase called OSAT – Outsourced Semiconductor Assembly and Test. It refers to a specific company model in the semiconductor supply chain that provide the third-party packaging and test services to chips [1]. Just like Fabless companies outsource to Foundries, now Foundries can also outsource this back-end stage to OSAT companies. In the “Design-to-Build” model, OSATs represent the back-end stage of production, whose responsibility for turning silicon wafers into the individual ready-to-use chips before reaching to customers. Many of the most famous tech giants follow this model such as Amkor Technology (USA) – leader in advanced packaging technologies; ASE Group (Taiwan) – the world’s largest OSAT service provider.

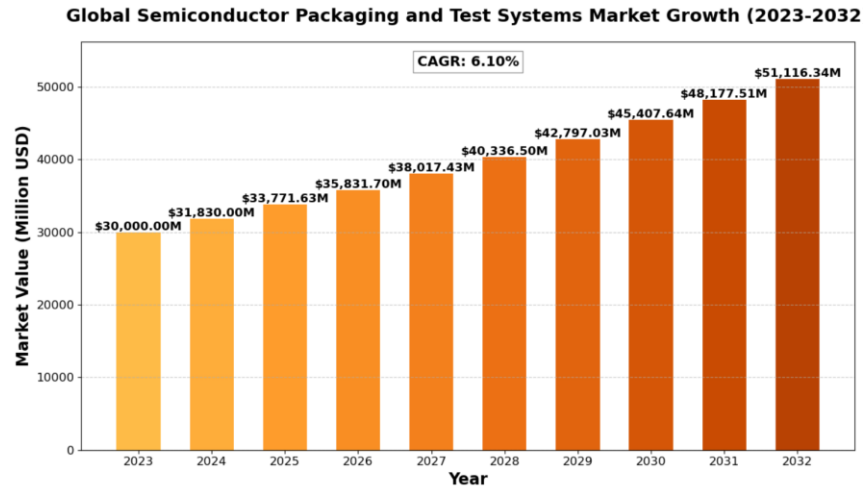


Fig 2. 1. *The global semiconductor packaging and test market growth (semiconductorinsight.com)*

As the figure above, the global semiconductor package & test market value is continuously increasing: from USD \$30000 million in 2023 to around USD \$51116 million in 2032, growing at a compound annual growth rate (CAGR) of 6.1% [2]. This growth indicates the increasing demand for the advanced semiconductor packaging and test solutions to meet the performance and reliability requirements.

2.1.2. The semiconductor testing process

The test stage is one of the most critical phases in the semiconductor manufacturing, defined as the process of measuring and verifying the electrical characteristics of a product after completing the fabrication stage [1]. In this project, we are intended to dive deep into the testing process following these main topics: classification, cleanroom and test equipment.

About the classification, the test process typically consists of two primary stages: Wafer Test (EDS Test) and Package Test (Final Test). First is the wafer test. It refers to the process that classifies whether individual chips meet all the specifications or not. The classification is achieved by measuring and verifying the electrical characteristics while chips are still in the wafer state, following the completion of the previous fabrication stage. This process is also known as Electrical Die Sorting (EDS) or Probe Test [1]. The test process typically involves

these following steps. First, the test conducts at a standard room temperature 25 °C, which is called the room temperature test. The test conducts at an extreme high and low temperatures, which is called the hot and cold test. The test conducts at a heat-stress temperature, used to identify chips that might fail early in their lifecycle, called the burn-in test [1]. After conducting these steps, any chips in the wafer which do not meet all the requirements will be identified as defectives. Defectives refer to the rejected products, which mark inked [1].

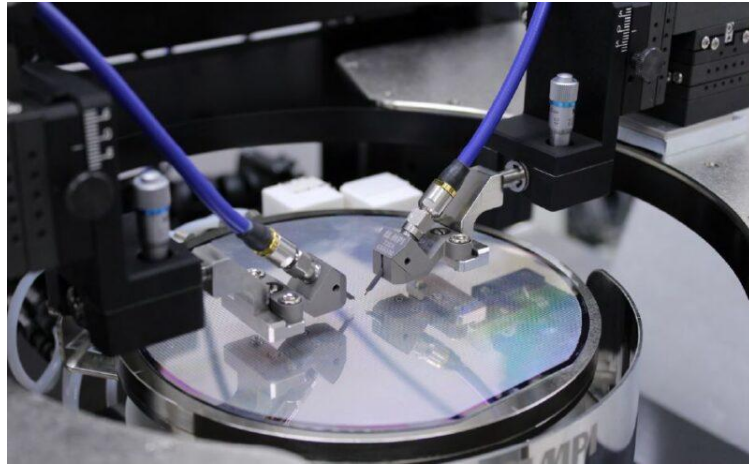


Fig 2. 2. The probe test reference (source: datatec.com)

Second is the package test. It refers to the process that classifies whether packaged chips meet all the specifications or not. The classification is achieved by measuring and verifying the electrical characteristics while chips are already finished the package process. This stage is also known as Final Test. Products will undergo some strict test phases such as the room temperature test at 25 °C; burn-in stress test under the harsh conditions; electrical and functional testing at various temperatures - 5 °C, 25 °C and 75 °C. after conducting these tests, any packaged chips which do not meet all the requirements will be identified as defectives [1]. Defectives refer to the rejected products, which are discarded. In contrast, the packaged chips pass all the test will be categorized into different bins based on their speed, functionality, ...



Fig 2. 3. The package test reference (source: tek.com)

About the cleanroom, the semiconductor manufacturing process is performed in a highly controlled environment that requires a high degree of cleanliness [3]. A space where dust particles, temperature, humidity, and air pressure are tightly controlled in this manner is called the cleanroom. The cleanroom plays a critical role in the reliability of the production process. Without operating in the cleanroom, dusts in the air fall onto the components during the manufacturing process, product defects can occur or the devices may fail to function [3]. Therefore, at semiconductor fabrication sites, factors like dust, temperature, humidity, and airflow are strictly controlled.



Fig 2. 4. ISO Certified Cleanroom (isocleanroomchina.com)

Whenever it comes to filtering system in cleanroom, the HEPA filter is the top of priority list. It continuously removes pollutants and contaminants from the atmosphere, helps improve air quality during the manufacturing process. Furthermore, to reduce sources of dust contamination from human factor, workers

must wear personal protective equipment (PPE), including white coat, gloves, booties, cap and mask while ensuring the entire body is covered except for the eyes. Before coming inside the cleanroom, workers need to go through the air shower system. Air shower removes the dirt and small particles from the surface of materials and body entering into the cleanroom. It is crucial to ensure that all workers follow PPE regulations to minimize contamination risks as much as possible.

The unit used to represent the cleanroom level is ISO Class (International Organization of Standardization) under ISO 14644-1, which is calculated based on the number and the size of particles per cubic meter of air (m^3). The lower the class number, the fewer the particles [3]. In the wafer fabrication process such as photo-lithography, wafer manufacturing and processing, diffusion process, ... cleanrooms are managed from Class 100 or Class 1 – the highest standard of cleanliness. In the meantime, the assembly and test phase, usually managed from Class 100000 to Class 1000 – the highly controlled environment [1]. See the following figures:

Class	Wafer manufacturing	Photo-lithography	Diffusion	Wafer processing	Assembly	Inspection
Office						
100000						
10000						
1000						
100						
10						
1						

Fig 2. 5. *Process in various classes level of cleanroom*
(source: LetuinEdu, *Hiểu về công đoạn chế tạo bán dẫn*)

About the test equipment, the test process and the equipment are inseparable. Without the specialized hardware, the test process can not be completed in terms of accuracy, reliability and speed. There are some equipments closely related to the objectives of the project, including the test instrument and the test fixture. First is the test instrument. In this project, we only focus on this precision semiconductor test instrument – The HP 4280A 1MHz C Meter / C-V Plotter.



Fig 2. 6. The HP 4280A 1MHz C Meter / C-V Plotter test instrument

It is a fully automatic, high performance test instrument designed to measure the C-V and C-t characteristics of semiconductor devices and materials [4]. The 4280A measures the capacitance of the device under test (DUT) as functions of constant dc bias, swept dc bias or time after application of a bias pulse [4]. In this project, we measure parameter C mode with a constant dc bias voltage supplied from the internal dc bias source. See the following table:

Table 2. 1. Typical specifications set before testing

No.	Specifications	Description
1	Measurement Parameter	Capacitance C (pF)
2	Measurement Mode	C mode
3	Operational Parameter	DC bias (DC V)
4	Trigger Source	Internal DC bias source = (dc bias) Output: - 50VDC Range: (-100V to +100V) Ranging modes: Auto (V LIMIT = 0) Resolution: 100mV Accuracy: $\pm(0.1\% + 0.1V)$

5	Connection Mode	Floating (CN10)
6	Measurement Terminal	Two-terminal pair configuration (HI and LO) with guard terminal.
7	Measurement Speed	MEAS SPEED: SLOW in C mode
8	Measurement Range	100pF/1mS
9	SIGNAL LEVEL	30mVrms $\pm 10\%$
10	Measurement Accuracy	$\pm(0.1\%$ of rdg + 3 counts)
11	Cable Length	1m
12	Test Signal	Frequency: 1MHz sine wave Accuracy: $\pm 0.01\%$
13	Error Correction	Enabled (key indicator is lit)
14	Self Test	Yes
15	Warm-up Time	≥ 30 minutes
16	Ambient Temperature	23 °C ± 5 °C

Cables, wires which used to connect to the test leads or the test fixture will have residual parasitic inductance that will cause an additive measurement error. So, the error correction function should be used to compensate for the residuals and strays of the cables [4]. Depending on [4], the Internal Error Correction procedures (1 meter test leads) are explained as following steps:

- (a) Connect one end of the test leads to the HIGH and LOW unknown terminals in the 4280A. The other ends connect to a test fixture.
- (b) Select 1m with the cable length key.
- (c) Connect nothing to the test sample end of the cables.
- (d) Set the instrument controls: connection mode, measurement mode, bias voltage, signal level, ... as appropriate for the desired measurement
- (e) Press ZERO OPEN button. Measurements will resume after about one to one and a half seconds.

- (f) Make sure the Enable of the Correction key indicator lamp is lit, ensure the error correction function is on.

Here is an example of the full cycle C mode measurement, depending on [4]:

- (1) Power on the instrument, warm-up at least 30 minutes
- (2) Select ∞ with the INTERNAL BIAS key
- (3) Key the desired number value. Press ENTER.
- (4) Turn the INT BIAS switch to determine voltage range. The desired range is 0V to ± 100 V for value – 50V. So, turn it up
- (5) Connect one end of the test leads to the HIGH and LOW unknown terminals in the 4280A. The other ends connect to a test fixture.
- (6) Turn on the instrument, press FLOATING connection mode.
- (7) Perform the internal error correction procedure.
- (8) Set the output voltage of the internal dc bias source.
- (9) Press the SET/RESET key to activate the internal dc bias source.
- (10) Read the C and G measurement display

Second is the test fixture. It is a specific connector where the DUT placed tightly in there to do the test. The test fixture serves as the physical and electrical interface between the Test Instrument and the DUT.



Fig 2. 7. The high-frequency test fixture reference

However, the test fixture always presents its parasitic capacitance by itself. Every physical mechanical structure made of conductors separated by insulators

such as air, plastic or through cabling, ... forms a parasitic gap. Metal contacts and pads facing each other across the air gap, results in creating parallel plate effect.

2.1.3. The existing issue

Measurement is the foundation of semiconductor testing. But not all times, the measurement between the test instrument 4280A and the fixture is not completely reliable and high-efficiency. Especially, in the pF range measurement level, the existing parasitic effects which always present in every conductive path act as a real matter, can degrade performance and lead to unreliable measurement result. Parasitic effects refer to unwanted parasitic capacitance existing in electronic circuits.

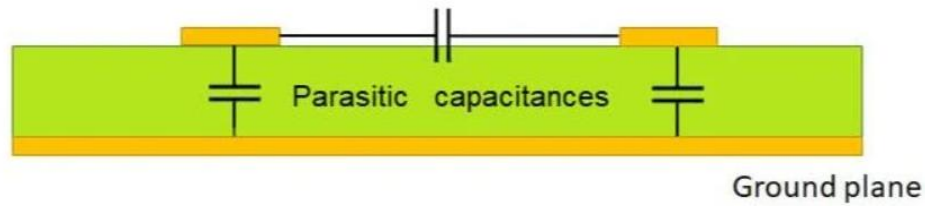


Fig 2. 8. Parasitic capacitance in PCB layout (source: vse.com)

Parasitic capacitance refers to unnecessary mutual capacitance that exists between the wiring, electronic components close to each other as parasitics. It is also known as the stray capacitance [5]. Generally, the capacitor consists of two conductive plates and insulating material (called a dielectric). According to the fundamental formula of the parallel plate capacitor:

$$C = \frac{\epsilon A}{d}$$

Equation 2. 1. *parallel plate capacitance formula*

Where:

C is capacitance, capacitance refers to value of the electric charge storage between conductors, measured in Farad (F)

A is the area between the two plates (m²)

d is the distance between the two plates (m)

ϵ_0 is constant of free space, an insulating material (appro. $8.85 \times 10^{-12} F/m$)

That formula shows a relationship: capacitance equals the surface area times the dielectric constant of the material, divided by the distance between the conductors. Because no insulator and no distance are perfectly fixed, everywhere can accidentally form into a capacitor, not just where we want it. That thing is a problem. For example, a circuit has many wires, the two parallel wires routing closely to each other can create unavoidable capacitance between two wires through air gap. So parasitic capacitance exists everywhere, presents anytime in PCB board design.

In low-frequency circuit, these effects may be ignored, but in high-frequency circuit, especially while measuring in the sensitive level range, these effects significantly matter [9]. Firstly, the crosstalk between signals. When two wires run parallel and close to each other, the existing parasitic effects create the unwanted signal from one wire to the other. If the digital switching signal goes fast and closer together, the situation gets worse to neighboring trace, leads to an unwanted voltage spike and data signal loss [6]. This phenomenon is called crosstalk.

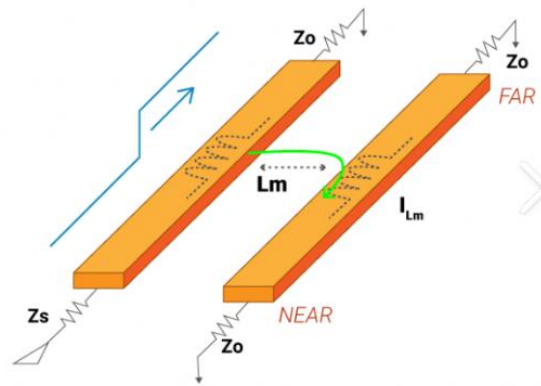


Fig 2. 9. Crosstalk in PCB design (protoexpress.com)

Secondly, unintended parasitic effects can lead to increase power consumption. The relationship can be explained by these simple equations:

$$E = \frac{1}{2} C_p V^2$$

Equation 2. 2. Electric field energy formula

Where:

E: electric field energy (W). The energy stored in a parasitic capacitor.

C_p : parasitic capacitance (F). the higher this value, the more unnecessary energy stored inside components.

V: voltage (V).

$$E = \frac{1}{2} L_p I^2$$

Equation 2. 3. Magnetic field energy formula

Where:

E: magnetic field energy (W). the energy stored in the magnetic field.

L_p : parasitic inductance (H). the higher this value, the more unnecessary energy created.

I: current flow (A)

Each time the signal switches, the more energy created in a unintended way for charging and discharging parasitic capacitors, the charging current flows through our components. This current generates heat, which can lead to increase thermal stress and power consumption [6].

Thirdly, unintended parasitic effects can lead to reduce switching speed and increase the propagation delay (t_{pd}). Propagation delay is the time it takes for a signal to travel from its source to its destination [7]. Average propagation delay time is calculated:

$$t_p = \frac{t_{pHL} + t_{pLH}}{2}$$

Equation 2. 4. Propagation delay formula

Where: t_p is the propagation delay time (s)

t_{pHL} is the time it takes the output to go from HIGH to LOW (s)

t_{pLH} is the time it takes the output to go from LOW to HIGH (s)

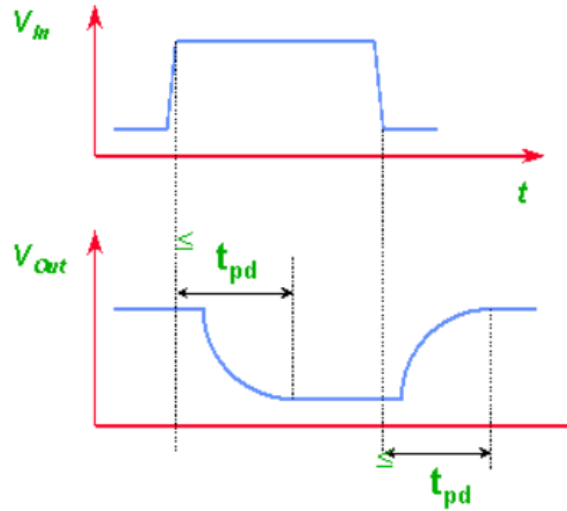


Fig 2. 10. Propagation delay in the switching signal (protoexpress.com)

In high-speed digital signal, each time microcontroller triggers switching signal between high signal (for example: 5V) and low signal (for example: 0V). every conductive path on PCB has a small amount of resistance. The larger parasitic capacitance in PCB, the longer switching time from 5V to 0V t_{pHL} / from 0V to 5V t_{pLH} requires. Instead of a sharp square wave, now the signal becomes a rounded curve (see the fig above.). At a result, due to parasitic capacitance, the signal requires more time to switch, leads to timing errors and slow switching speed.

2.1.4. The approaching method

The main causes of increasing this parasitics come from several aspects of PCB design, including improper PCB layout rules, components selection, wiring and traces, ground planes, interconnects and also one thing - physically manual cable/probe switching [8]. In this section, the proposed approach to find the solutions for minimizing noise effects is discussed.

Firstly, switch box implementation for solving manual switching problem. This is also the objective of the project. Switch box, is a device used to manage and control the switching flow signals between input sources and output destinations. In various fields, especially in high-accuracy measurement in industrial laboratory and factory, the switch box device plays a vital role in ensuring reliability, high accuracy. Instead of manual switching by physical

plugging/unplugging cables in risks of inconsistent data readings and potential damage, now switch box offers a great solution to switch data sources effortlessly and flexibly.

Secondly, components selection method before implementation. There are many types of components which have different operating specifications. For example, some resistors are designed to be low inductance, some transistor for low package inductance and fast switching [8]. In our noise-suppressed demand, we should choose the components that help minimize the parasitic effects as much as possible.

Thirdly, PCB layout techniques for noise optimization. There are many basic techniques to handle such as separating power and signal traces through the partition method based on functional behavior, the integral shielding method, routing wire trace lengths, ... [9].

Another solution we approach is the high-controlled environment. Technically, the testing experimentation needs to be carried out in a specially equipped lab, called the cleanroom, as the environment we mentioned above. Some factors such as temperature, humidity, airflow, ... needs to be strictly controlled to prevent their negative impact on our testing.

2.2. COMMON SWITCH ARCHITECTURES

2.2.1. Overview of the switch architecture

Based on the approaching method, we now dive deep into the fundamentals of switching solution. Generally, working with many automated test instruments in the industrial field often requires routing signals to a variety of instruments and device under tests. To handle this application, the best way is to implement switching system that keeps routing signal in control and increases the flexibility of our measurement in semiconductor testing. There are four main types of switching architectures: no switching, switching in test rack only, switching in test fixture only and switching in test rack and test fixture [10].

2.2.2. No switching

In this first architecture, the signal between the test instrument and the test fixture together with the devices under test is direct. It means no switching in the middle of measurement system. We call this architecture is “no switching”, or “direct connection”.

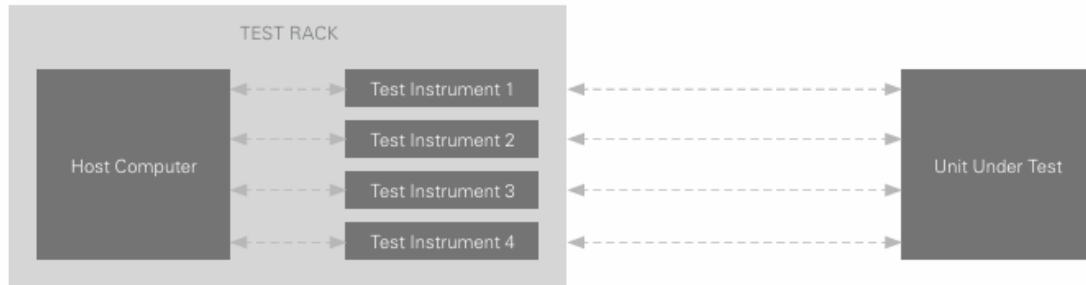


Fig 2. 11. No Switching architecture (source: NI.com)

If we use this architecture in our measurement between the test instrument HP 4280A and the test fixture, there will be no switching relays, no additional wires or multiple connectors in the measurement path. The signal now becomes pure as much as possible because it is absolutely minimized the parasitic noise effects, provides the accurate low-level results while measuring at sensitive level range [10]. Moreover, the throughput is excellent. Throughput refers to signal speed and bandwidth. Because the signal does not have to pass through a switching relay system and additional wire length, it reaches to the instrument instantly.

However, while the strengths of “no switching” seem like to be excellent, this architecture is not suitable for manufacturing phase. Manufacturing phase requires mass testing production over thousands of tests and repeatability of measurements. We can not manually unplug and replug connectors one by one while running the mass testing production. This architecture is only recommended to be used for R&D (Research and Development) laboratory or maintenance of machine.

2.2.3. Switching in Test Rack only

Second is switching in test rack only. Test rack refers to a central frame that used to house multiple test instruments, power supplies and host computer in a single cabinet. Usually, this cabinet type is commercial off-the-shelf, means that it is provided and sold as a switch product by the vendor [10].

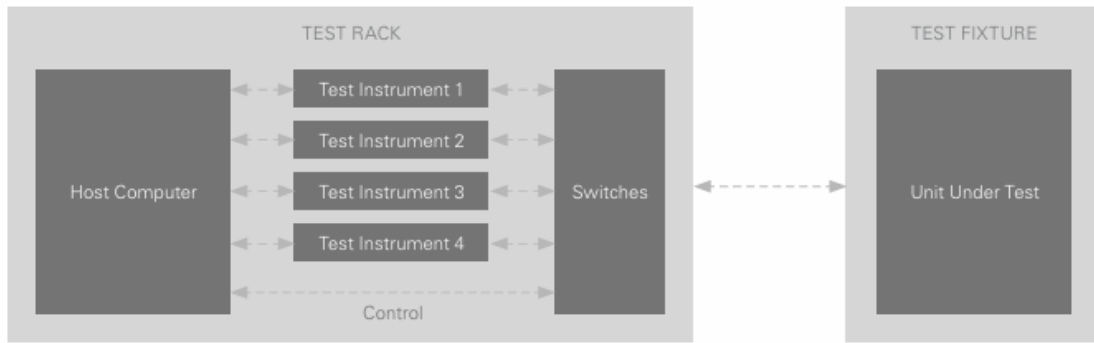


Fig 2. 12. Switching in Test Rack only (source: NI.com)

By using this architecture, we can save considerable development time, no need to spend time implementing PCB design and driver board for the switch system by ourself. The system can be scalable by purchasing additional types of switch modules from the vendor rather than redesign a new own thing, make this architecture to be the most flexible system to the expansion options [10].

Due to the high payment for this switch architecture, the cost is the main disadvantage needs to be considered. Technically, placing all of the switching within the single test rack can also increase the total amount of cable wire lengths, leads to increase the usage of cables between switches and test instrument, increase the usage of cables between the switches and test fixture. It is not suitable for the objectives of the project, which are limited to implement a switch box design in the economical way.

2.2.4. Switching in Test Fixture only

Third is switching in test fixture only. In this case, multiple switching relays are placed on a PCB design near the test fixture or inside the fixture itself. When using this architecture, we need a relay driver PCB board to control the individual relays respectively to the terminals of test fixture. Technically, using this architecture requires additional design work to drive switching relays.

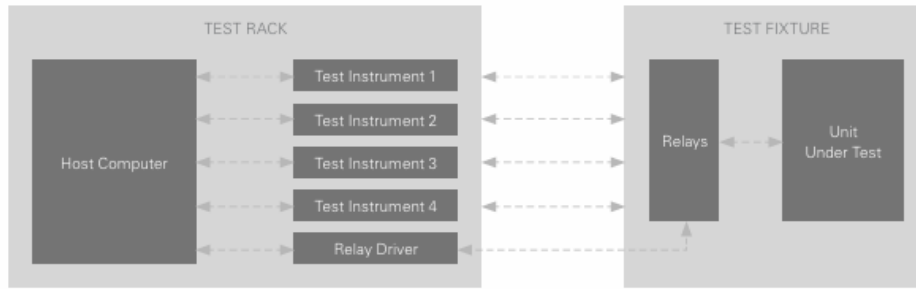


Fig 2. 13. Switching relays inside the test fixture only (source: NI.com)

Building the custom switching system helps reduce test cost and reduce the unwanted need for cables between the test instrument and the test fixture. Reduction in cabling results in minimizing the parasitic effects of the testing measurement, decreasing measurement error readings [10]. By localizing the relay driver to the respective switching relay at the fixture level, the signal path is considerably shortened, effectively reducing parasitic effects compared to other architecture. This is the main objective of this project we want to follow.

As mentioned above, we can not just use the relays alone, we need to build the relay driver to control our relays. This may add complexity to hardware design work. Building our own custom switching design can be an existing challenge to anyone who is lack of PCB design and technical experience before. Also, the fixed PCB board switching design disables the ability to scale the test system in the future.

2.2.5. Switching in Test Rack and Test Fixture

The last is the combination between switching in Test Rack and switching in Test fixture in the single architecture. Based on its characteristics, it can be called as a “Hybrid Switching” architecture. The application of this complex architecture usually used for “many-to-many” switching solution [10]. It means that a massive amount of test instruments and a massive number of test points in the test fixture requires, where one single switching device is not enough.

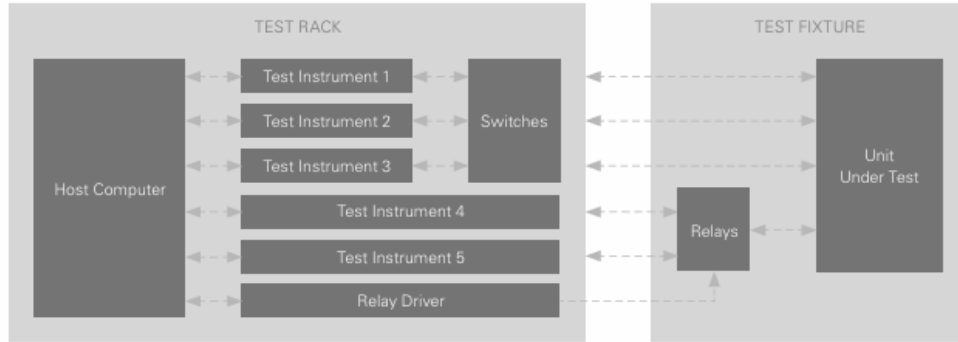


Fig 2. 14. Hybrid Switching (source: NI.com)

This architecture takes advantage of both two architecture as mentioned previously, switching in test rack only and switching in test fixture only. For instance, the switch system can be scalable by off-the-shelf advanced switches and simultaneously minimize parasitic effects in sensitive low-level measurements through custom relays in the test fixture [10].

Due to the highest complexity of this architecture, the cost is extremely expensive. Building this large-scaled systems can take long development time and require a significant technical PCB design expertise. For the project with a test instrument HP4280A and a specific diode test fixture, this architecture would be not suitable to be chosen. See the following table:

Table 2. 2. The trade-off in the choice of four switching architectures

Type	Flexibility	Accuracy Measurement	Cost	Repeatability	Time development	Result
No switching	Low	High	Low	Low	Low	Not suitable
Switching in Test Rack	High	Medium	High	Medium	Low	Not suitable
Switching in Test Fixture	Medium	Medium	Medium	Medium	Medium	Suitable
Switching in Test Rack and Fixture	High	High	High	High	High	Not suitable

In summary, based on the comparison of these four switching architectures, we choose the switching in test fixture as the main architecture for the project. The reason is that it provides the optimal balance between measurement quality and the flexibility. About the flexibility, the test systems are often built for specific chipset testing phase, so they are not usually expanded too fast. So, we should keep the medium level for flexibility factor. We don't need flexibility to be "super high" as Test Rack while we require noise optimization to be "medium above". The trade-offs are conducted carefully to choose "switching in the test fixture only" as the best suitable architecture for this project.

2.3. COMMON SWITCH TOPOLOGIES

2.3.1. Overview of the switch topology

In electrical-electronic device field, the switch topology is the logic arrangement of switching devices to ensure that the switching system meets efficiency and high-performance requirements. In terms of automated test

applications, the switch topology refers to the relay organization to create different types of relay network. A relay is a remotely controlled device that enables to make or break a connection in an electric circuit. Because of its usefulness, it is used in almost circuits, from the simple turn on/off the light bulb to the complex circuit in industrial automated system. It can be formed into many switch topologies such as multiplexer parallel level / cascade level, matrix and other hybrid topologies. All these types will be explained and compared the pros and cons to choose the best suitable switch topology for our project.

2.3.2. General-Purpose Relays

General-purpose relays are relays designed to meet a broad set of specifications that can work in many different environments. For example, they usually handle common voltages such as 12V, 24VDC or 110VAC, 220VAC, moderate currents typically 5A to 15A. In spite of being specialized in one advanced task, they are used for many universal applications. When we simply want to make/break a connection within a circuit or terminal, or turn on /off a light bulb or controlling a fan, a general-purpose relay is a great option.

There are three types of relays: single-pole single-throw (SPST) relay, single-pole double-throw (SPDT) relay and double-pole single-throw (DPST) relay [10]. SPST relay is a simple switch with on and off states. SPST relays come in two forms: normally open (Form A) and normally closed (Form B). Form A relays are normally open, only activates when the relay contact touches other connection to complete the circuit. Form B relays are normally close, only activates when the relay contact break the connection to open the circuit, see the following figure.



Fig 2. 15. Single-pole single-throw relay – SPST relay (source: NI.com)

A single-pole double-throw (SPDT) relay has a single pole and two connections, that can alternate between one normally open (NO) and one normally closed (NC). SPDT relay also comes in two forms: form C and form D.



Fig 2. 16. Single-pole double-throw relay – SPDT relay (source: NI.com)

A double-pole single-throw (DPST) relay is the combination of two form A SPST relays working simultaneously. We can build this relay from two independently controlled Form A SPST relays, usually work in the same coil and package together.

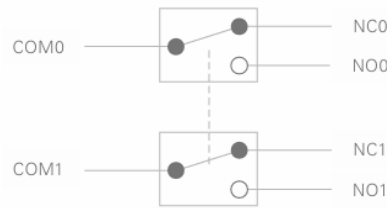


Fig 2. 17. Double-pole single-throw relay – DPST relay (source: NI.com)

2.3.3. Multiplexers

A multiplexer, or MUX, is a switch topology that enables to control relay from multiple inputs to one output. In the testing process, this is one of the most common topologies used for signal routing, which allows to connect any one of various different test points to a single terminal of the test instrument.

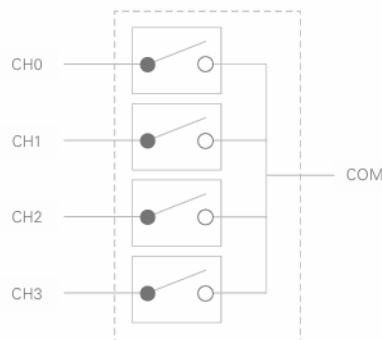


Fig 2. 18. The 4 x 1 MUX of parallel level SPST relays (source: NI.com)

To build a multiplexer, there are two common simple ways: parallel level and cascade level. Parallel level type is simple by using multiple form A SPST relays put parallel together with one end point connected, see the fig above, 4 x 1 MUX, for example. However, one disadvantage is that the unused signal paths in this parallel topology can do negative impact on the bandwidth of the switching signal, create AC signal reflections.

Another type is cascade level, using form C SPDT relays, see this fig below.

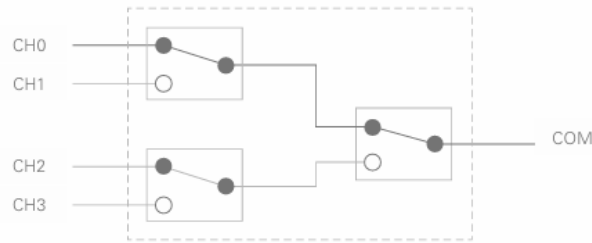


Fig 2. 19. The 4 x 1 MUX cascade level SPDT relays (source: NI.com)

This cascade topology exists to solve the parallel drawback. It can improve the signal integrity of AC signals because every switch relays in system is used functionally. However, this type often requires more PCB spaces.

2.3.4. Matrix

A matrix refers to an effective organizing way to switch multiple relays. It enables to control multiple signal paths at the same time based on rows and columns definition at each intersection. Matrix allows us to maximize the signal flexibility of the system when routing signals [10]. Matrix size is often defined as M rows by N columns (M x N), usually size 4x4, 16x16 and 4x5, see the following figure for example.

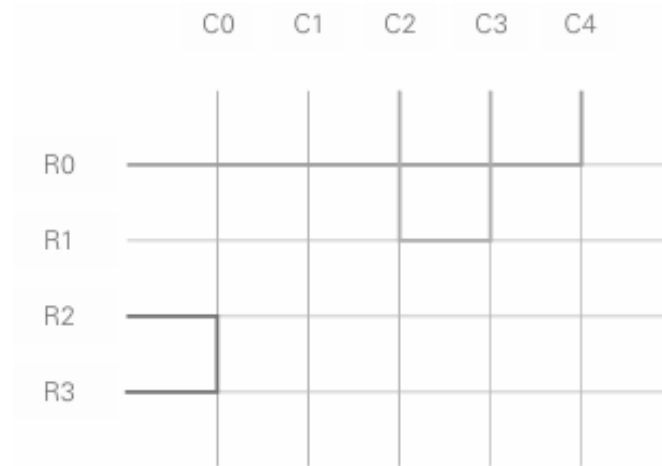


Fig 2. 20. The 4x5 matrix topology in switching system (source: NI.com)

The advantage of matrix topology is the maximum flexibility. We can connect any input to any output, depending on the matrix size (M x N). While a MUX topology is like a single path with many entrances, a matrix is now like a multi-lane highway path. A matrix allows signal to define unique one path. This is helpful while measuring between two test points, for example, the test instrument point and the test fixture point. However, due to only one unique path, the other unused rows and columns can introduce a high risk of parasitic effect. These stubs act as an open wire, lead to add potential noise effects.

CHAPTER 3. SYSTEM DESIGN

3.1. SYSTEM DESIGN

3.1.1. System requirements

Firstly, the switching system must be compatible with the test instrument and the test fixture. In details, to compatible with the test fixture and the test instrument, the hardware design needs BNC terminal for input, output function. Coaxial BNC connector is used to deliver the test signal with controlled impedance, helps minimize parasitic capacitance from cabling. Moreover, it shields the signal path from noise, especially efficient when measuring at the sensitive level range, see the following figure:



Fig 3. 1.BNC connector in the switching system

Secondly, the switching system should be designed in the two-floor architecture design. This architecture can help optimize well-suited size for convenient measurement to collect the capacitance characteristics of the test fixture. Because of the objectives of the project, the hardware needs to design to be optimized size for connecting and measuring to the test devices. Indeed, we choose to design the switching box to be in square shape. The size PCB board is accepted around 100 x 100 x 2 (Width x Length x Thickness, measured in mm value).

Thirdly, the switching system design must apply the theoretical fundamentals which have been explored in the noise-suppressed solution. In details, the choice of the components using in the PCB, the choice of the suitable switching architecture, switch topology and the strategic PCB routing layout principles are necessary methods to suppress parasitic effects in the switching system.

3.1.2. Block diagram

Based on the hardware requirements, a general block diagram of the switch box system is established, consisting of these following blocks:

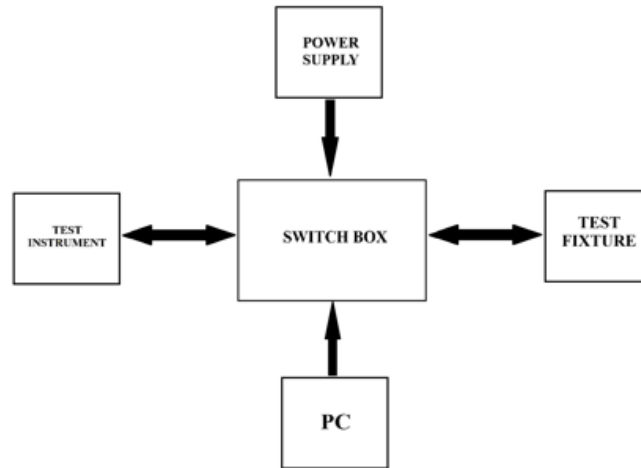


Fig 3. 2. Overall block diagram of the system

Where:

Firstly, the power supply block. This block is responsible for providing the main power needed for the internal components of the system. It ensures that the supplied voltage is stable, noise-filtered and compatible with other components.

Secondly, the PC block. this block is responsible for sending the input embedded code to the MCU of the switch box, controlling IO signal to switch relay.

Thirdly, the Test Instrument block. This block is responsible for measuring, collecting consistent reading to get test values. The test values demonstrate in the screen of the test instrument. In this project, the test instrument used names The Hewlett – Packard Model 4280A 1MHz C Meter/C-V Plotter. “It is a high-performance test instrument designed to measure the C-V and C-t characteristics of semiconductor devices and materials”.

Fourly, the Switch box block. This block represents as the main objective needs to implement in this project.

Finally, the Test Fixture block. This block represents as the connector where we place the packaged devices. Test fixture already haves its parasitic capacitance by itself. The objective is to minimize this noise at the Test Fixture block.

The detailed block diagram of the switch box is established:

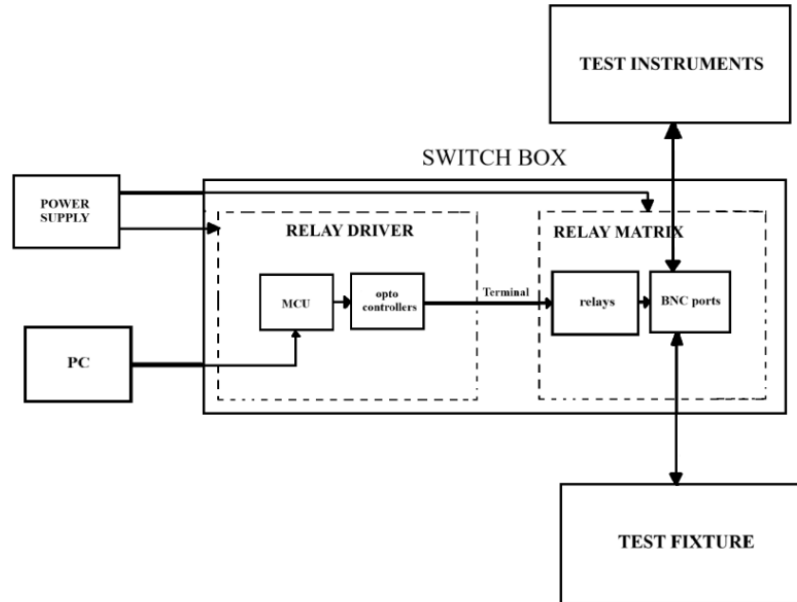


Fig 3. 3 .Detailed block diagram of the switch box

Following the hardware requirement, the power supply for internal two main boards of the switch box is separated. The two main boards are linked together through the DB25 terminal. The relay driver board: consists of MCU and opto-controllers. Each relay is controlled by each opto-controller, so we will use 12 opto-controllers to control 12 relays. The relay matrix board consists of relays and BNC ports.

3.2. HARDWARE DESIGN

3.2.1. Requirements for hardware design

Based on the figure, the hardware implementation must meet these following requirements:

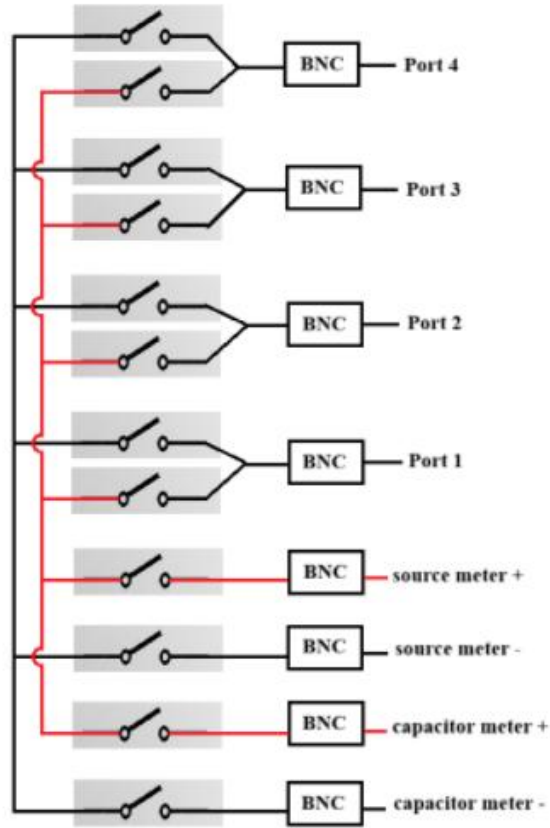


Fig 3. 4. *Hardware design requirements*

Firstly, depending on the above fundamentals of building a switch system, we choose the most suitable switch architecture for the project is the switching in test fixture only. Also, the general-purpose relays matrix topology is used for this project.

Secondly, the hardware needs to consist of two main PCB boards: relay matrix board and relay driver board. Relay matrix board is formed by 12 relays which are suitable for noise optimization. Relay driver board is formed by microcontroller unit (MCU) and 12 opto-controllers which works as a relay driver. Each relay is driven by each opto-controller. Two this main boards form into a switch box.

Thirdly, the switch box is supplied by 5V DC power voltage, where: the power supply source for relay matrix board and the power supply source for relay driver board should be isolated. It means that each board use its own different source power to prevent potential conflicts which can make unwanted noises.

3.2.2. Schematic diagram

The switch box consists of two main internal boards, each corresponding to its specific function: the relay driver circuit and the relay matrix circuit. Here, the

schematic diagram is designed by EasyEDA. First, the relay driver circuit is described as following:

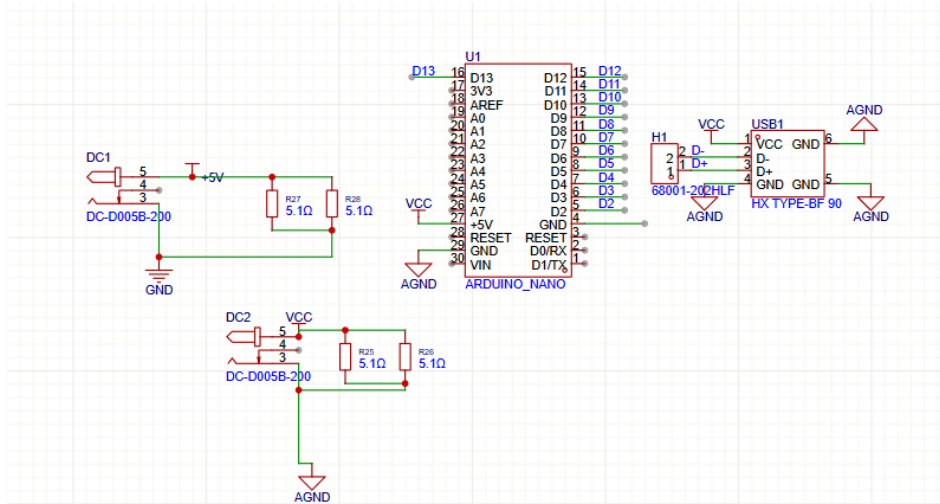


Fig 3. 5. Schematic diagram of MCU circuit in the relay driver board

We choose MCU ATmega328P, 8-bit 16 MHz clock speed with memory 32 KB Flash, 2 KB SRAM. This is the core processing unit which is responsible for handling most of the system's critical tasks.

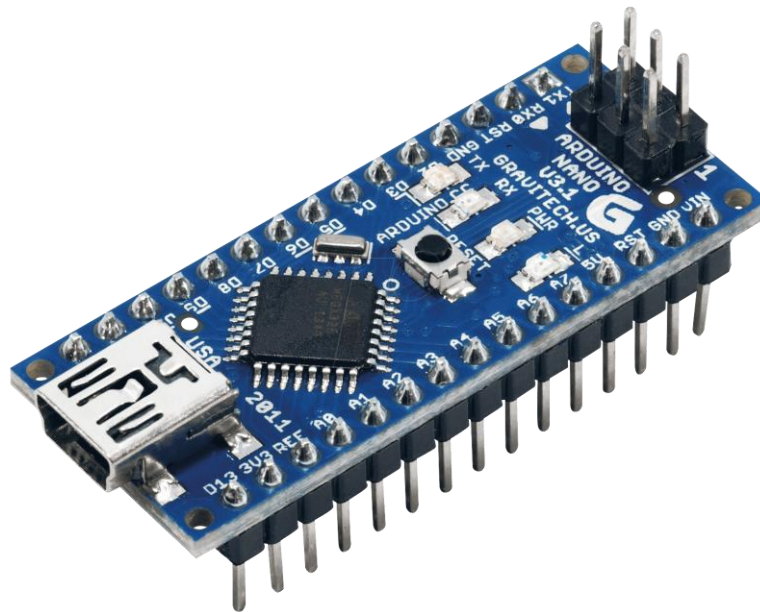


Fig 3. 6. MCU ATmega328P Arduino Nano

The Arduino Nano offers the same core functionality as the Arduino Uno but in a much smaller form factor, making it ideal for space-constrained projects. It offers a rich set of GPIOs and communication interfaces, which are useful for connecting external components. Arduino Nano has strong community support and

is compatible with Arduino IDE, which provides easy development through many valuable libraries.

Table 3. 1. The MCU Arduino Nano pins connector

Pins	Function
VCC	Power voltage +5V
GND	Connect to GND
D2	Control relay1 port P4+
D3	Control relay2 port P4-
D4	Control relay3 port P3+
D5	Control relay4 port P3-
D6	Control relay5 port P2+
D7	Control relay6 port P2-
D8	Control relay7 port P1+
D9	Control relay8 port P1-
D10	Control relay9_cappos port S+
D11	Control relay10_capneg port S-
D12	Control relay11_srcpos port C+
D13	Control relay12_srcneg port C-

The Arduino Nano microcontroller operates at range 3.3V to 5V DC and has Mini-B USB connector for programming. For the easy approach, we choose power supply adapter 220VAC to 5VDC. This device is known for widespead availabilty and cost-effectiveness.

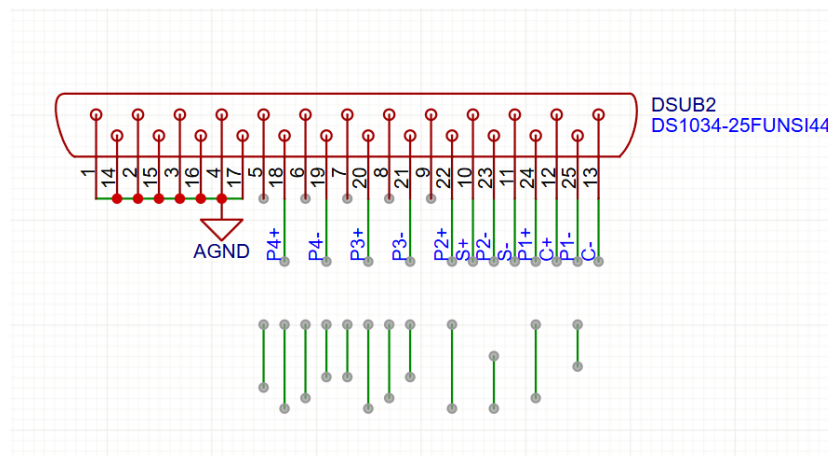


Fig 3. 7. Schematic diagram of relay driver board connection pins



Fig 3. 8. *DB25 male and female connector*

DB25 connector, serial type: DS1034 is used for communication from the top driver relay board and the bottom relay matrix board. See the following DB25 connector pin table:

Table 3. 2. DB25 connector pins

No.	Pin	Function
1	1,2,3,4,14,15,16,17	Connect together to the same common GND
2	18	Used for P4+ signal
3	19	Used for P4- signal
4	20	Used for P3+ signal
5	21	Used for P3- signal
6	22	Used for P2+ signal
7	23	Used for P2- signal
8	24	Used for P1+ signal
9	25	Used for P1- signal
10	10	Used for S+ signal
11	11	Used for S- signal
12	12	Used for C+ signal
13	13	Used for C- signal

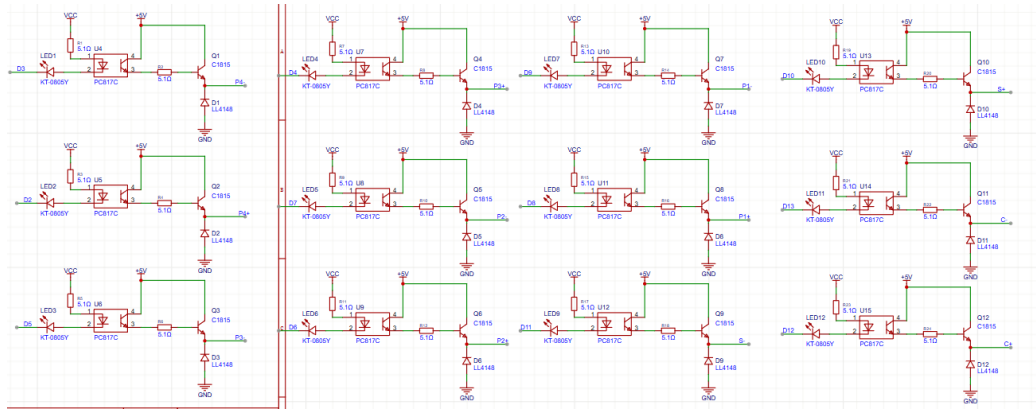


Fig 3. 9. Overall schematic diagram of 12 opto-controllers circuit

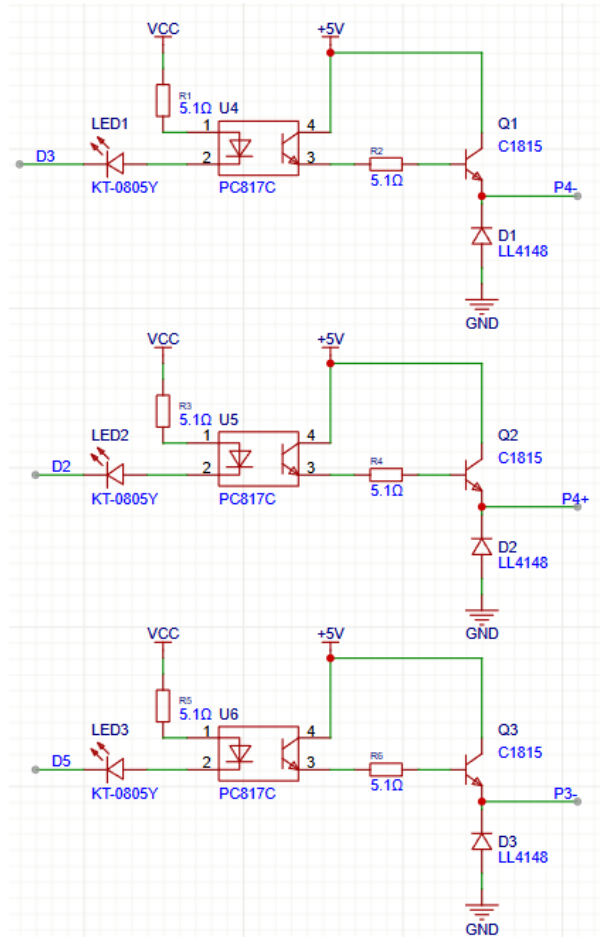


Fig 3. 10. Detailed schematic diagram of opto-controller circuit

We choose photocoupler model PC817C for opto-controller circuit. It serves as a bridge between the sensitive low-voltage signal and the high-power relay switching signal. The PC817C can handle voltage at collector-emitter up to 35V.



Fig 3. 11. *PC817C optocoupler (source: PC817C datasheet)*

Standard MCU pins provide 5VDC, the PC817C Led needs around 10mA to 20mA, so we calculate:

$$R_{led} = \frac{V_{cc} - V_{led_opto} - V_{led_red}}{I_{led}}$$

Equation 3. 1. *Led resistane Formula*

$$= \frac{5V - 1.2V - 2V}{10mA} = 180\Omega$$

Hence, we can choose current limiting resistor 0603 type value $220\Omega \pm 1\%$.
The transistor base resistor 0603 type value $1k\Omega \pm 1\%$

When the input signal goes low, the LED turns on, which results in turning on the output photo-transistor to drive the base of the C1815 transistor. For safety, we include flyback diodes LL4148 connected from the emitter of the transistor C1815 to ground. The purpose is to protect transistors from the inductive voltage spikes when the relay turns off. For more details, see the pin number and internal connection at the following figure:

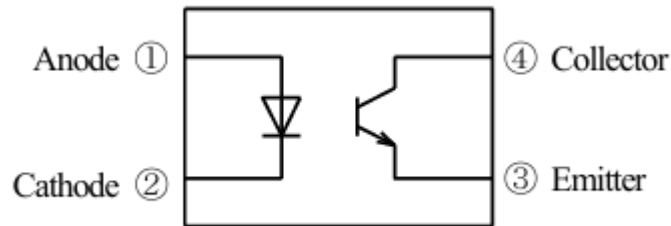


Fig 3. 12. *PC817C pin diagram (source: PC817C datasheet)*

Second, the relay matrix circuit is described as following:

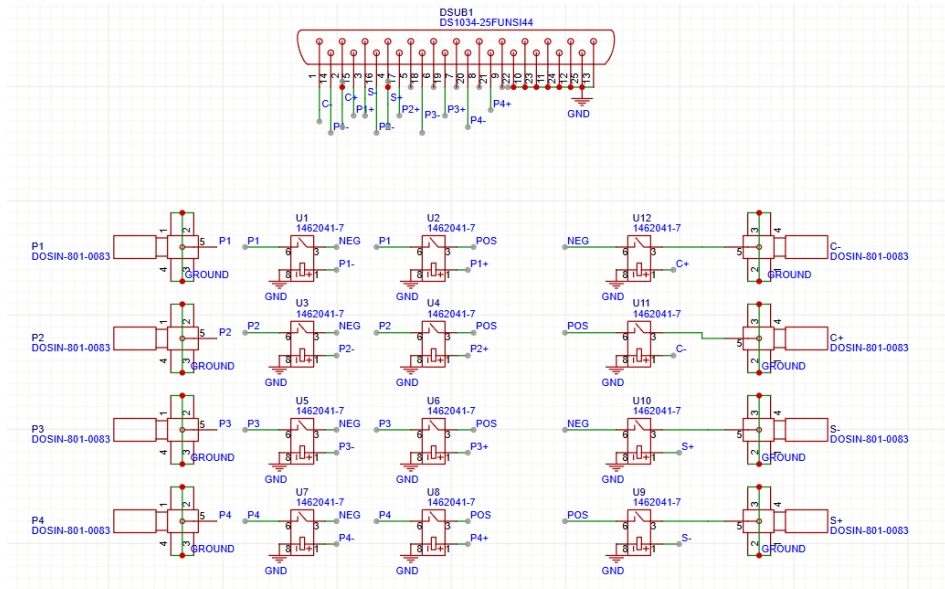


Fig 3. 13. Overall schematic diagram of 12 relays matrix circuit

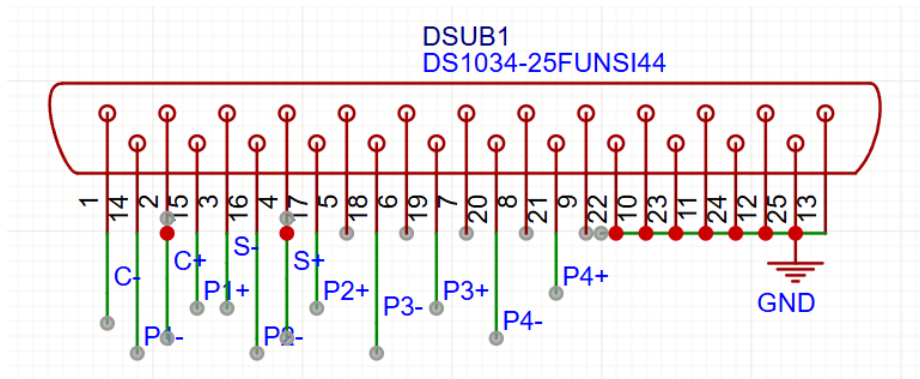


Fig 3. 14. Schematic diagram of relay matrix board connection pins

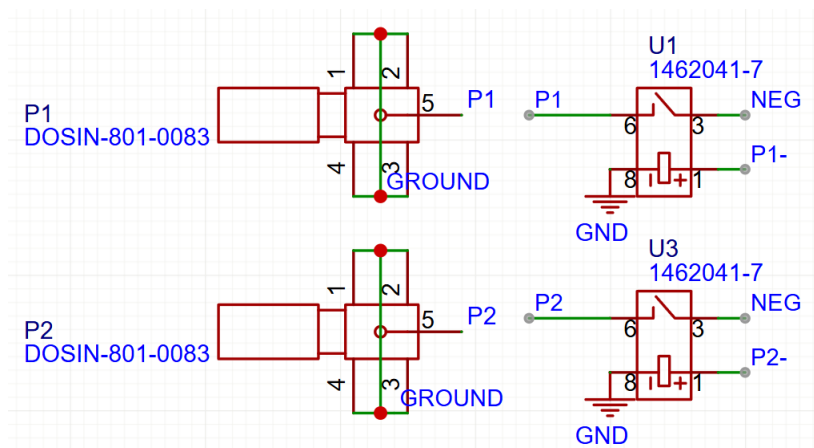


Fig 3. 15. Detailed schematic diagram of the relay matrix circuit

The relay chosen to be used in the relay matrix circuit is an Axicom IM03C signal relay, manufactured by TE Connectivity. This is a general-purpose signal

relay type, normally open type, 2 form C DPDT-CO, 5V DC coil voltage, 250V AC contact voltage.

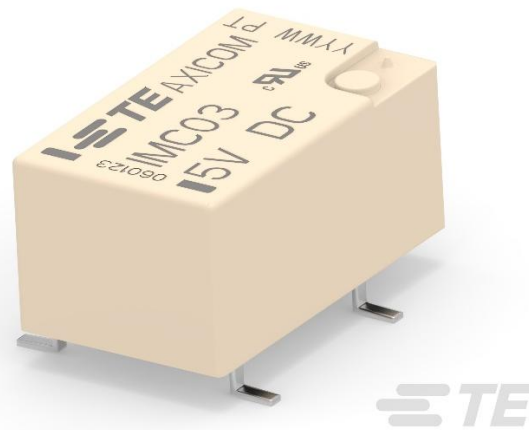


Fig 3. 16. *IM03CGR relay (source: TE Axicom)*

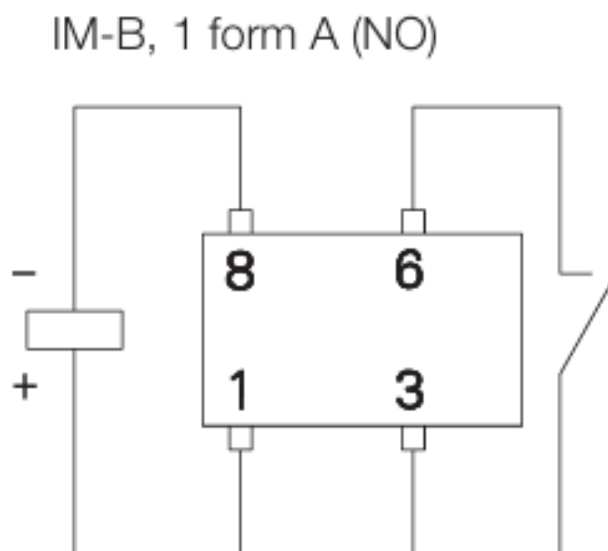


Fig 3. 17. *Pin diagram of the IM03CGR relay (source: IM03CGR datasheet)*

Let's see the pin diagram above as an example, the relay block will activate when the pin number 1 triggers HIGH, Pin 1 is the positive side of the coil, and Pin 8 is always connected to ground. When we apply a HIGH signal to Pin 1, the current flow through the coil, creating a magnetic field that pulls the internal switch contact from Pin 6 to Pin 3. At the result, the signal from Pin 6 to Pin 3 is connected. Because this relay type is normally open (NO), so the connection from pin 6 to pin 3 will be disconnected if there is no current flow through the coil.

The low coil power consumption is designed for high-performance switching relay applications. Using this specific signal relay is a great choice for low-cost and compact signal switching. Because it is a signal relay rather than a high-power industrial relay, it generates less noises, which helps minimize noise in the hardware design.



Fig 3. 18. *BNC connector structure (metabee.com)*

The BNC connector used in the relay matrix circuit is DOSIN BNC connector. The BNC connector 50 Ω impedance type is primarily used for radio frequency applications, telecommunications and high-performance test equipments [11]. It is crucial for achieving accurate measurements and ensuring reliable performance while testing with the test instrument [11].

3.2.3. Layout PCB

After completing the schematic design, we proceed to PCB layout in Easy EDA tool. Here is the following observation:

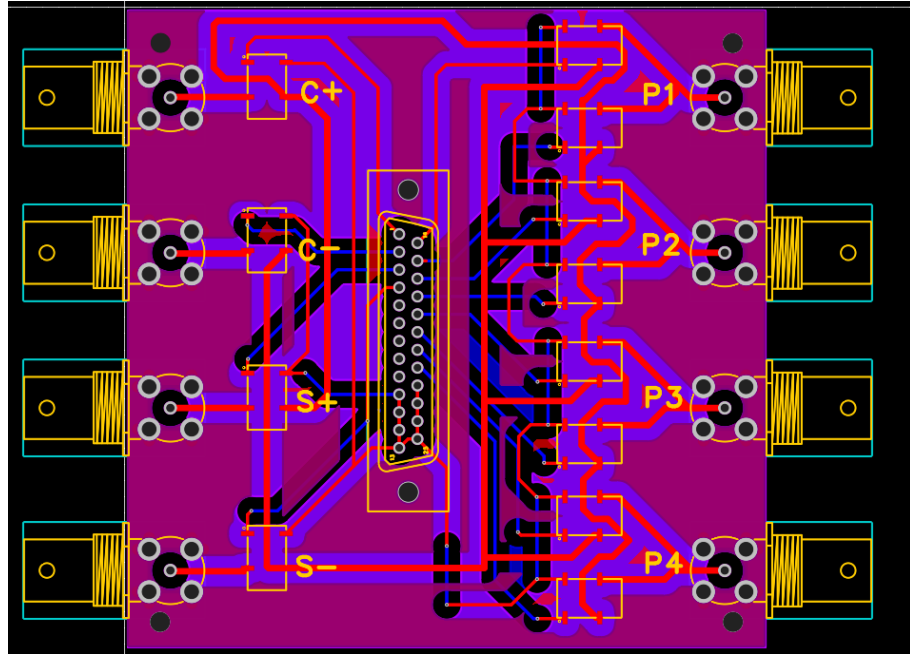


Fig 3. 19. *Top layer PCB layout for the relay matrix board*

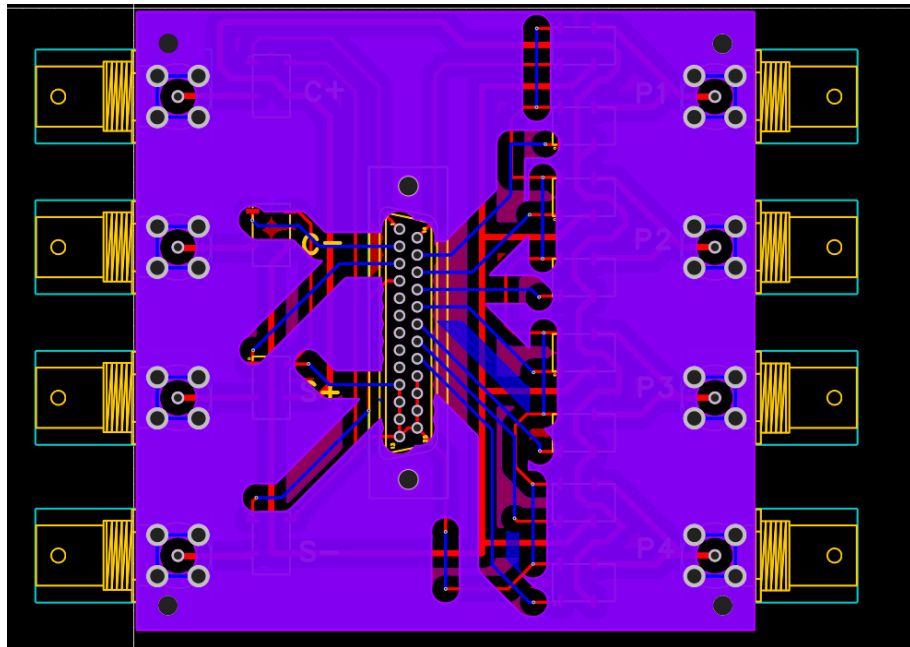


Fig 3. 20. *Bottom layer PCB layout for the relay matrix board*

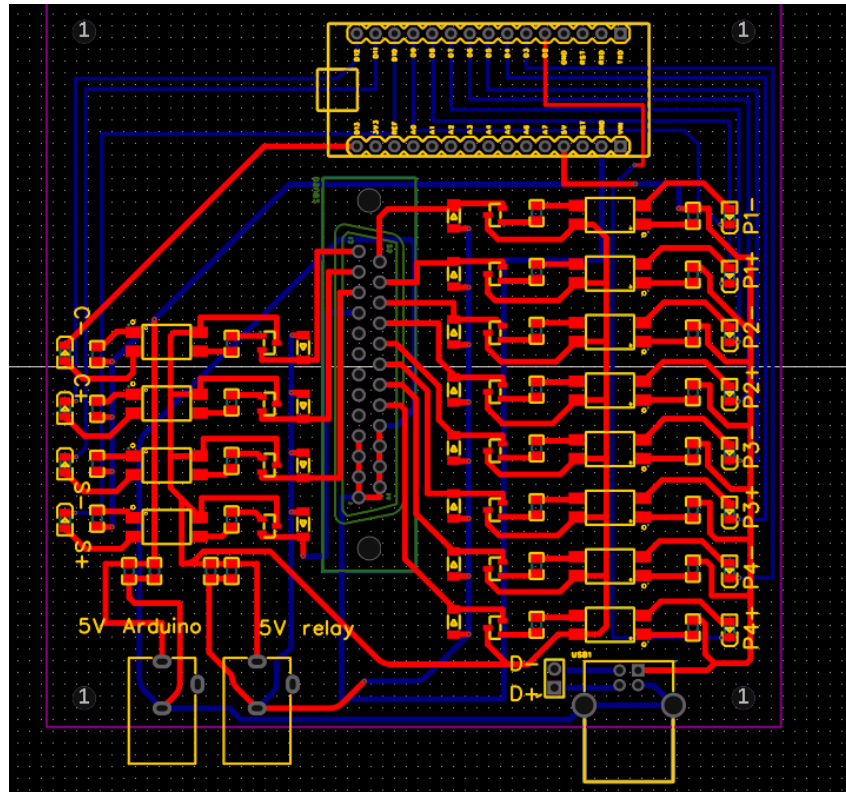


Fig 3. 21. *Top layer PCB layout for the relay driver board*

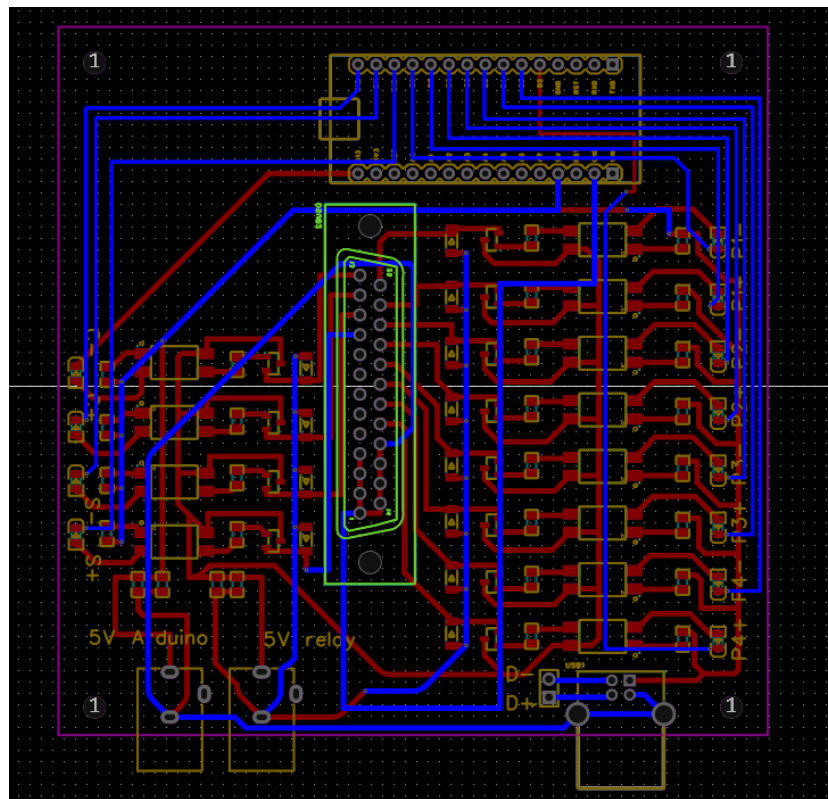


Fig 3. 22. *Bottom layer PCB layout for the relay driver board*

Parasitic effects are unavoidable in high-frequency PCB design, but with some strategic methods, their impact can be minimized. Here is explained as below:

About the partition, we should separate functional zones in PCB to prevent noise coupling. For example, we arrange and place the microcontroller unit and many digital signal components such as PC817C optocouplers, capacitors, C1815 transistors into the relay driver board, called the digital zone. In the meantime, we place BNC connectors and relay contacts into the same zone, called the analog zone. This zone is the most sensitive area, needs to be separated. Two of these zones only communicate through the DB 25 pins serial port.

About the wiring and traces, we use the Electroless Nickel and Immersion Gold (ENIG) plating technique in the relay matrix PCB layout. This technique uses the gold-plated copper surrounding traces, which is usually applied in high-frequency circuit design. This shield works as a “Faraday shield” to prevent the cross-talk for several long parallel traces. For example, see the figure of the relay matrix PCB layout above. We can see “S+” signal and “S-” signal very closely parallel to each other, results in potential leaks. But thanks to ENIG technique, we keep a wide clearance between the trace and the gold pour, the channel-to-channel interference is suppressed.

About the choice of passive components, we choose ceramic capacitors with low equivalent series inductance (ESL) for decoupling. Hence, current limiting resistor 0603 type value $220\ \Omega \pm 1\%$. The transistor base resistor 0603 type value $1\text{k}\ \Omega \pm 1\%$ is chosen. Inductors with low equivalent series resistance (ESR). Use surface-mount resistors instead of through-hole resistors with low inductance. Active components: transistors, diodes, ICs. Choose components with low package inductance and capacitance. The flyback diode LL4148 is a great choice because it is a fast-switching diode with low package capacitance. The relay Axicom IM03C 5VDC coil voltage is a great choice. The BNC connector $50\ \Omega$ impedance type is chosen.

About the interconnects and vias, we minimize the use of vias on the critical signal paths by keeping the signals mostly on the top layer to maintain signal integrity. Solder joints needs to ensure cleanness, minimal solder joints by using the right amount of solder paste during assembly.

About the size of PCB, the driver relay board is designed as 100 x 100 x 2 (Width x Height x Thickness, in milimet value), with tolerance of ± 0.5 mm . Then, the relay matrix board is designed in the same size as the driver relay board: 100 x 100 x 2 (Width x Height x Thickness, in milimet value), with tolerance of ± 0.5 mm. The distance between two boards is 23mm (± 0.5 mm).

3.3. SOFTWARE IMPLEMENTATION

3.3.1. Requirements for software implementation

Firstly, each MCU Arduino Nano pin must be clearly defined and control each relay respectively. In details, we need to define GPIO , which pins of MCU uses to control which specific relays. See the pin definition in the following table: Table 3. 3. Pin definition of MCU

No. Pin of MCU	No. Port channel	No. Relay
D2	port P4+	RY1
D3	port P4-	RY2
D4	port P3+	RY3
D5	port P3-	RY4
D6	port P2+	RY5
D7	port P2-	RY6
D8	port P1+	RY7
D9	port P1-	RY8
D10	port S+	RY SRCPOS
D11	port S-	RY SRCNEG
D12	port C+	RY CAPPOS
D13	port C-	RY_CAPNEG

Secondly, while running logic code, each specific input serial string should be defined for each different functional operation. See the following table for detailed explanation:

Table 3. 4. The input serial string definition

No.	Input serial string	Function
1	CONNECT_CAP	Drive {RY_CAPPOS, RY_CAPNEG} to low, Drive {RY_SRCPOS, RY_SRCNEG} to high
2	CONNECT_SRC	Drive {RY_CAPPOS, RY_CAPNEG} to high, Drive {RY_SRCPOS, RY_SRCNEG} to low
3	PORT1_CH+	Drive {RY7} to low
4	PORT1_CH-	Drive {RY8} to low
5	PORT2_CH+	Drive {RY5} to low
6	PORT2_CH-	Drive {RY6} to low
7	PORT3_CH+	Drive {RY3} to low
8	PORT3_CH-	Drive {RY4} to low
9	PORT4_CH+	Drive {RY1} to low
10	PORT4_CH-	Drive {RY2} to low
11	CLEAR_RELAY	Drive {RY1, RY2, RY3, RY4, RY5, RY6, RY7, RY8, RY9, RY_CAPPOS, RY_CAPNEG, RY_SRCPOS, RY_SRCNEG} to high

Thirdly, when power-up code execution, the MCU must immediately drive all relay control pins to a known OFF state to eliminate floating signals. The reason is to prevent a mismatch where the MCU thinks a relay is open, but it is actually closed due to a previous power cycle.

3.3.2. Flowchart

The system flowchart consists of multiple logic segments including:

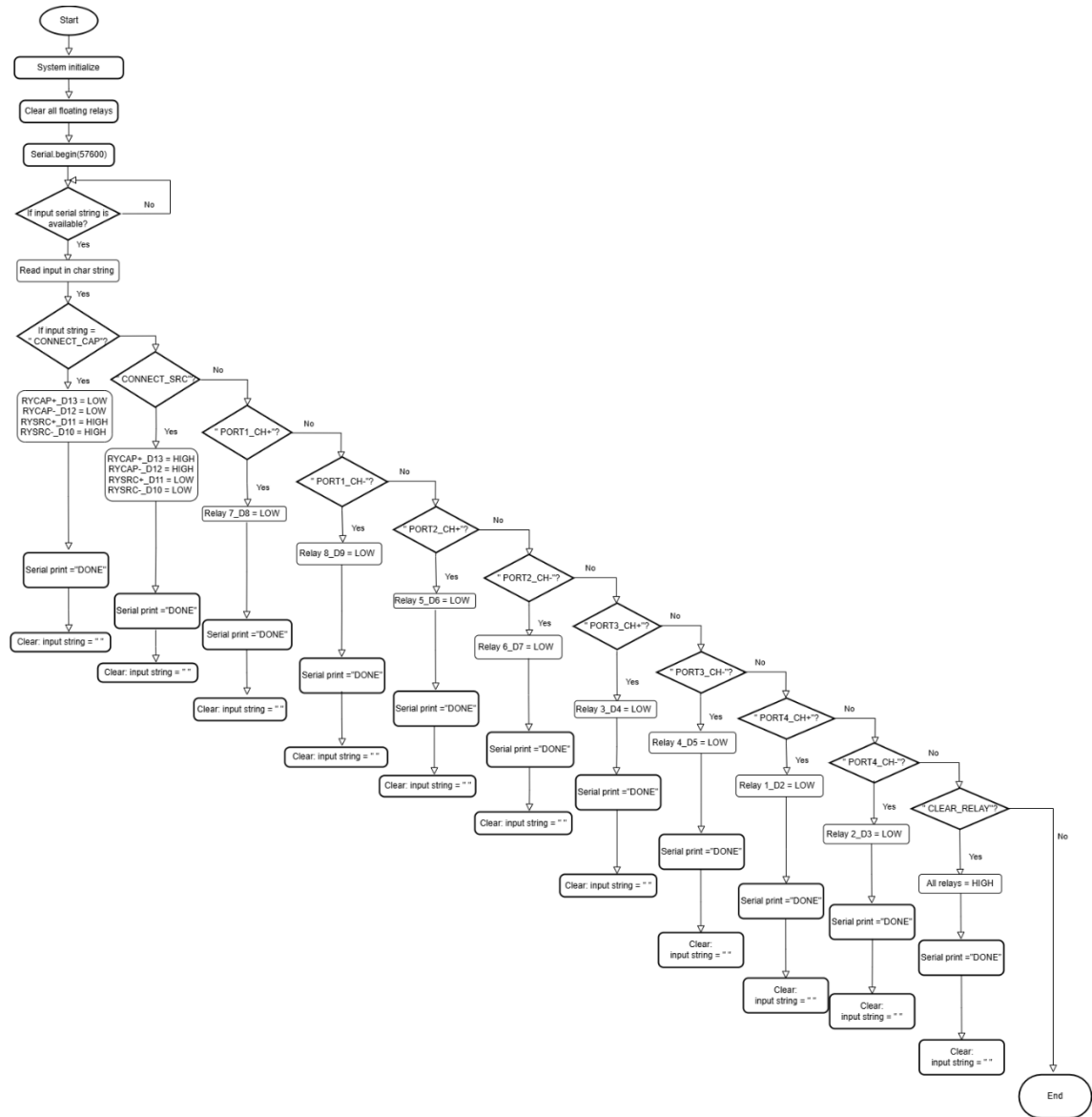


Fig 3. 23. Flowchart of the switching relay system

After power supply, the system first performs the initialization. This process includes: MCU board detection, communication port detection and GPIO pin definition. In logic code, using `#define` to map the MCU's digital pins to specific relays. Following the hardware requirements, we need to define 12 relays, where 2 relays for each 4 ports, totally 8 relays to connect with the test fixture. The other is 4 relays to connect with the test instrument HP 4280A. the logic code organizes the matrix 12 relays into two distinct groups: Selection relays from Relay₁ to Relay₈, to choose which port (from P1 to P4) and which polarity (+ or -) is active. Function relays from CAP to SRC to choose which the measurement mode (Capacitance or Source) of the HP 4280A.

CHAPTER 4. RESULTS

4.1. THE OUTCOME OF THE PROJECT

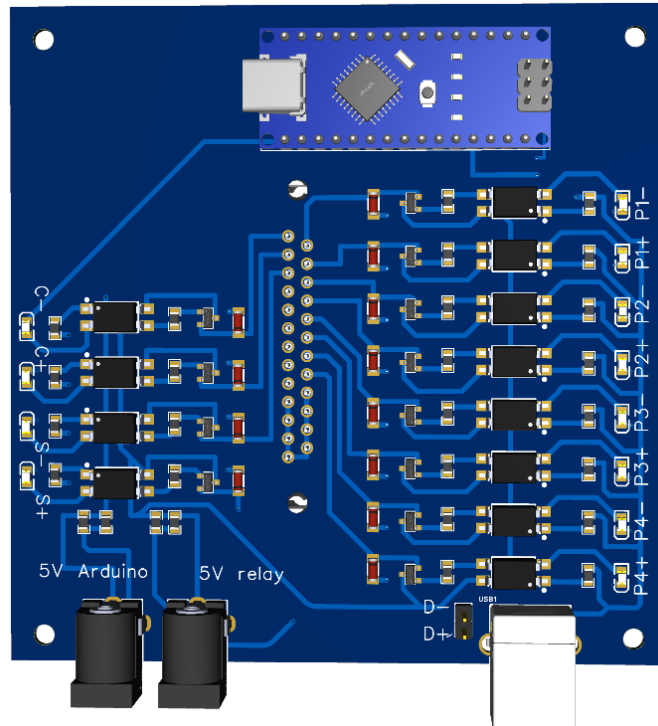


Fig 4. 1. *Top layer 3D view of the driver relay board*

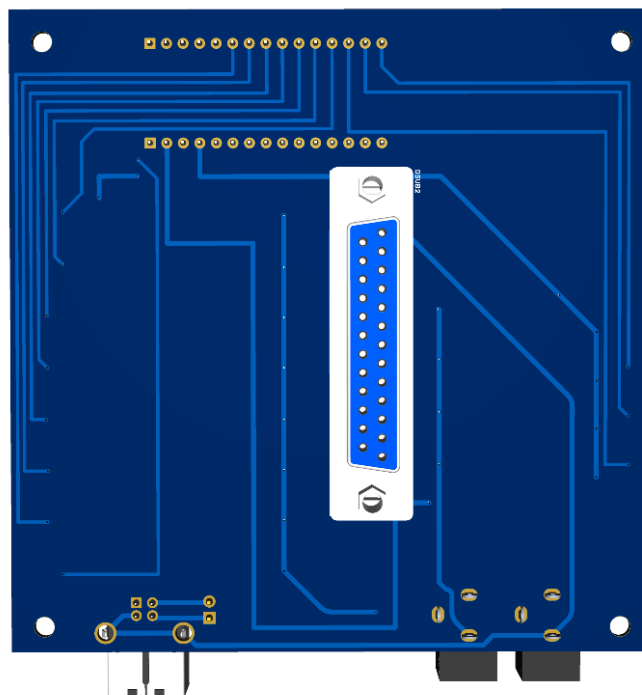


Fig 4. 2. *Bottom layer 3D view of the driver relay board*

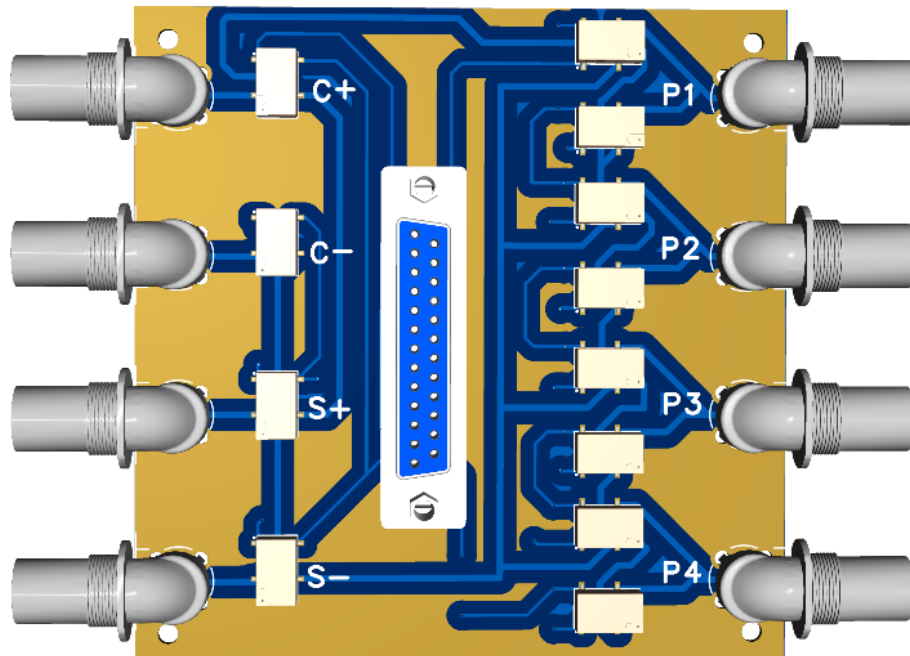


Fig 4. 3. *Top layer 3D view of the relay matrix board*

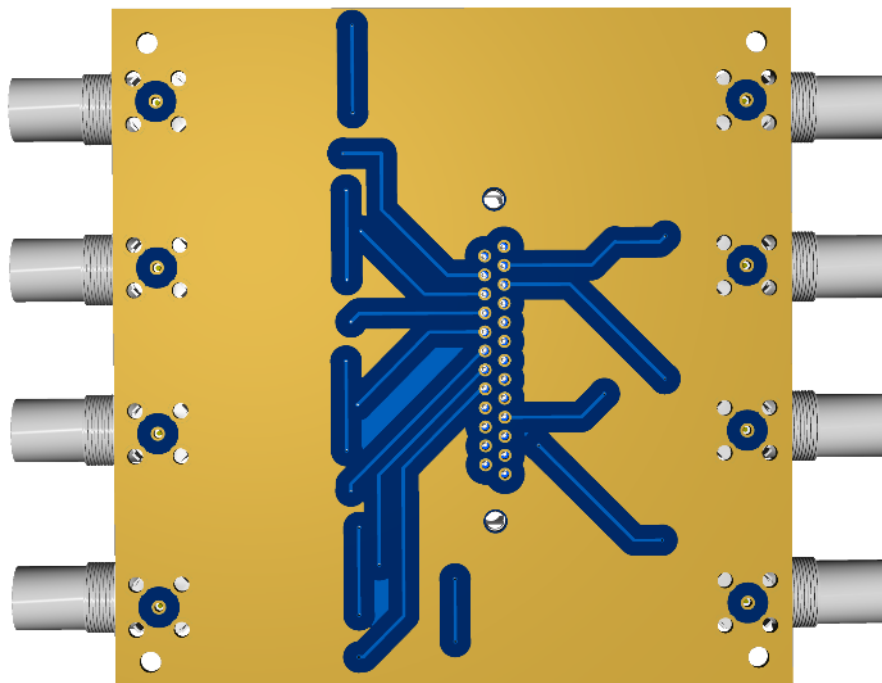


Fig 4. 4. *Bottom layer 3D view of the relay matrix board*

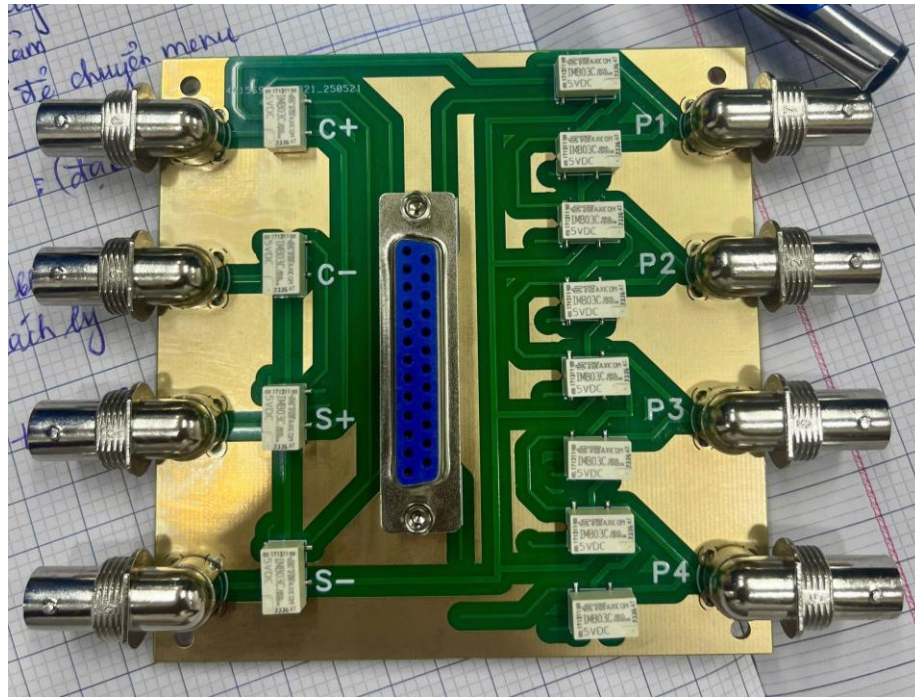


Fig 4. 5. *The front side of the relay matrix board*

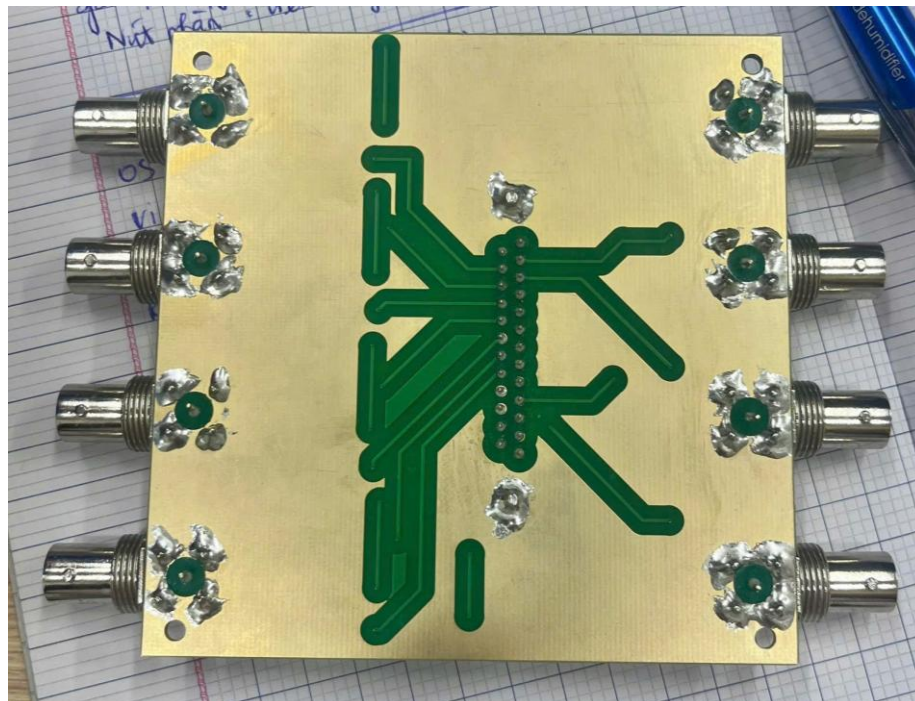


Fig 4. 6. *The behind side of the relay matrix board*

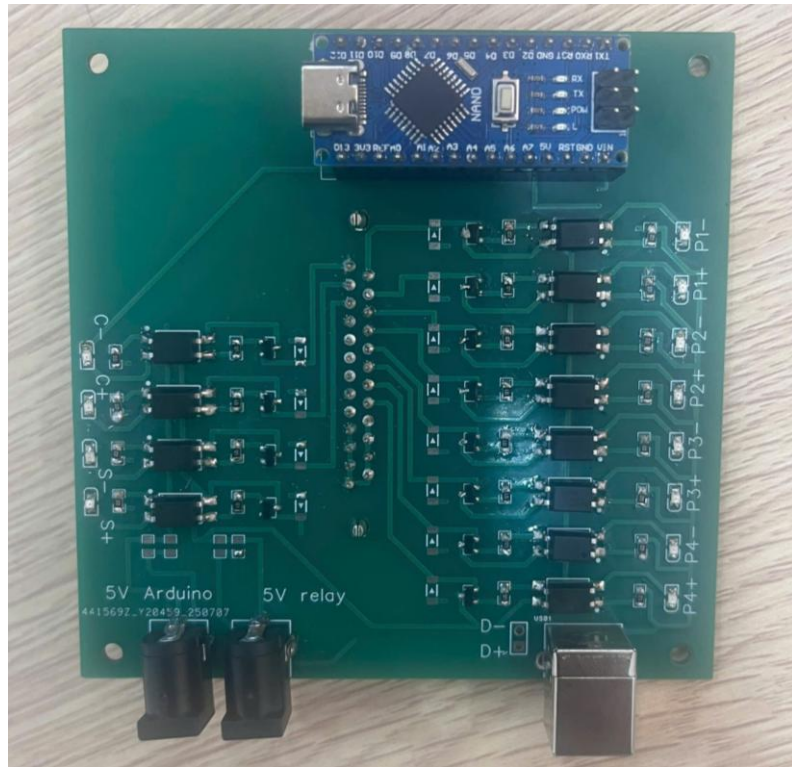


Fig 4. 7. *The front side of the relay driver board*

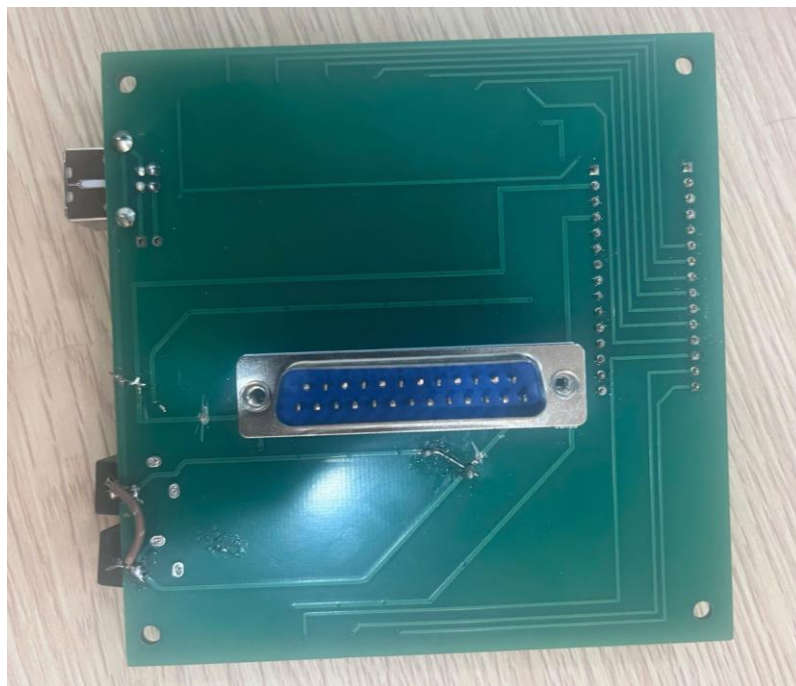


Fig 4. 8. *The behind side of the relay driver board*

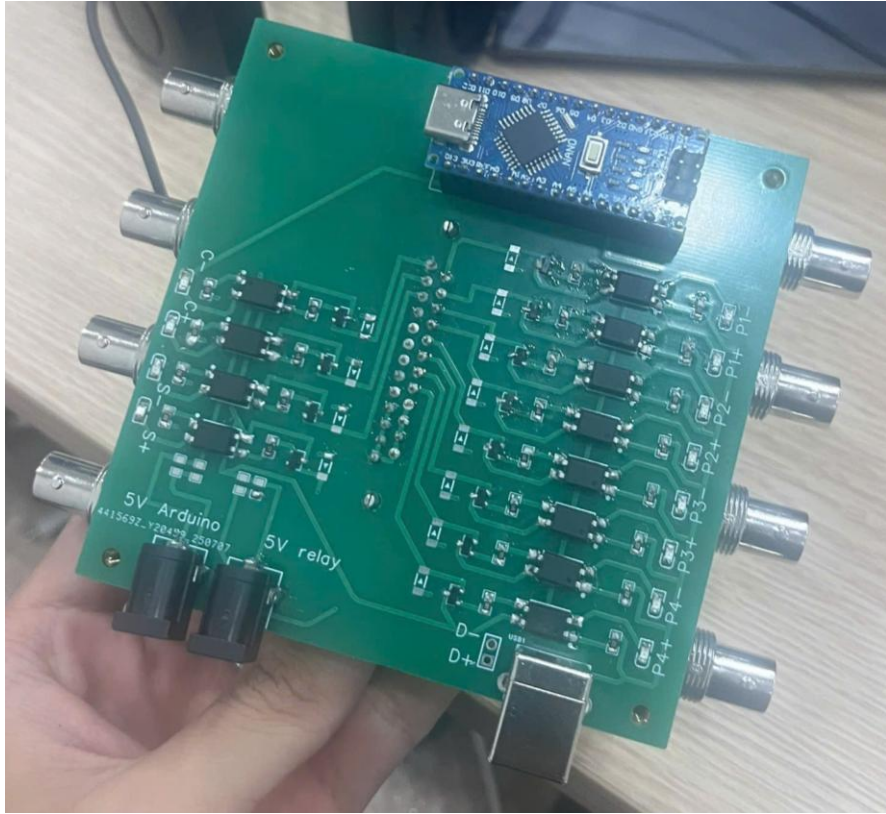


Fig 4. 9. *The top side of the switch box*

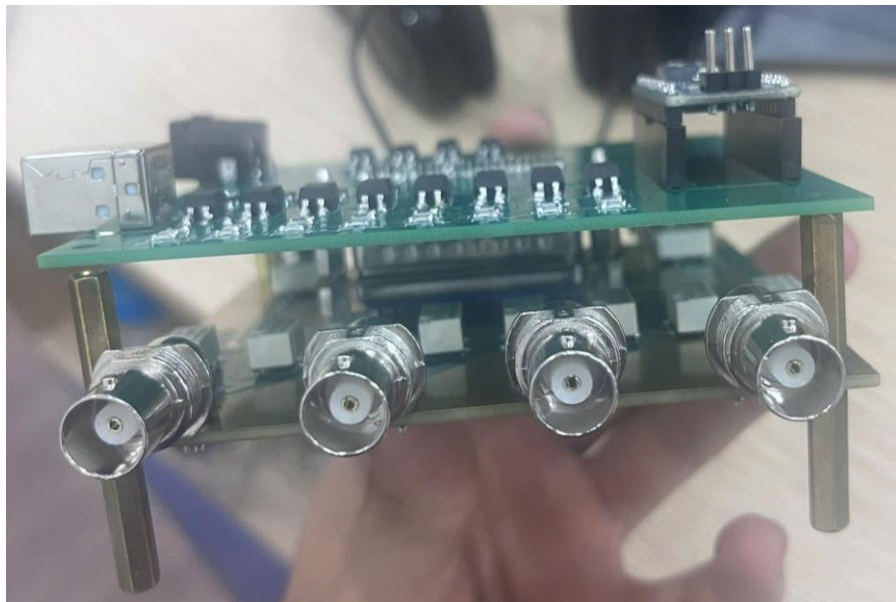


Fig 4. 10. *The left side of the switch box*

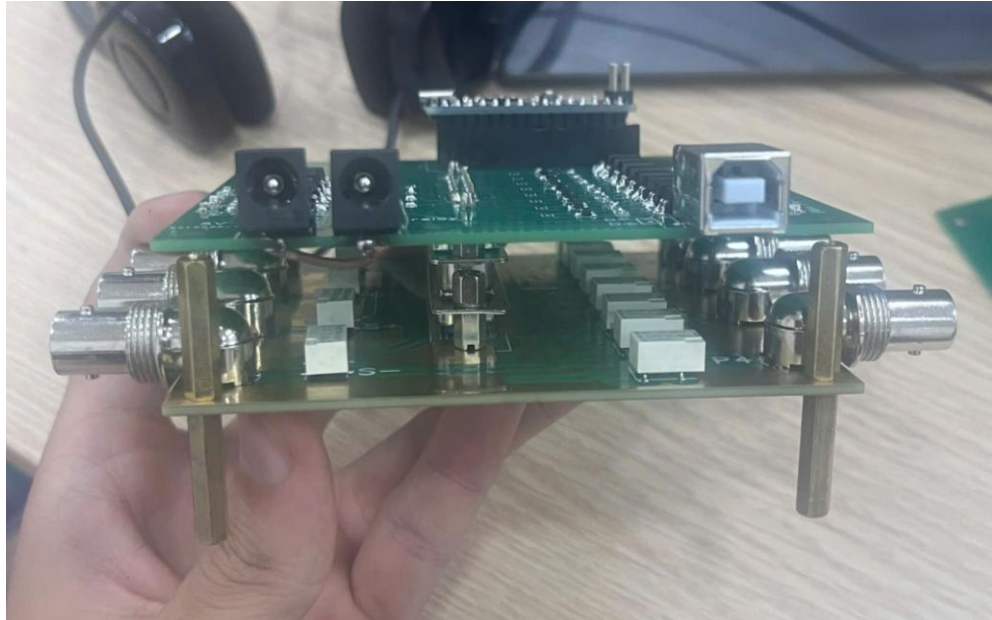


Fig 4. 11. *The side position of the switch box*

4.2. DEMO

Here is the two-phase testing plan. First phase is the functional verification. The goal here is to ensure the switch box work exactly switching relays according to logic code. Powering up the switch box, open the Arduino IDE Serial Monitor, set to 57600 baud and send input string commands. For example, “CONNECT CAP” to turn on two relays for two signal “C+” and “C-”, see the figure below:

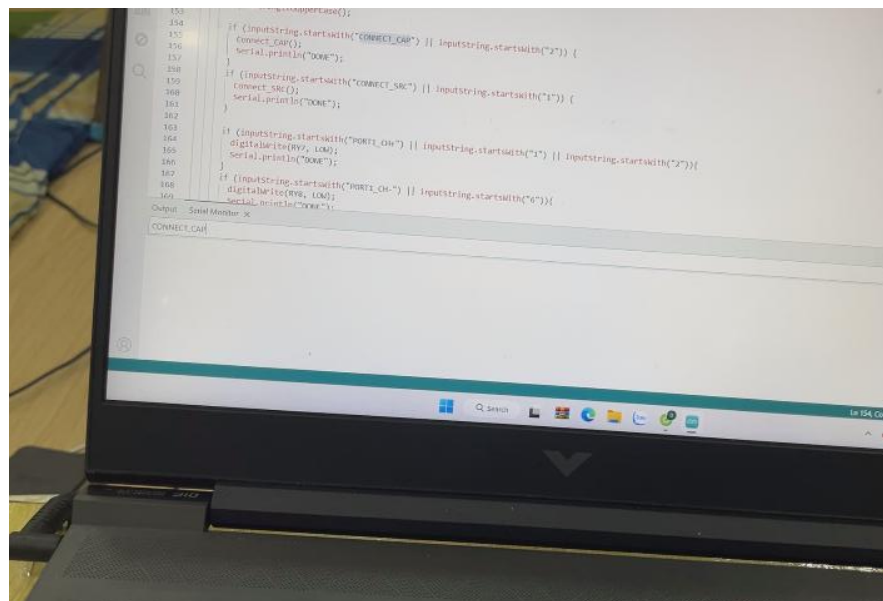


Fig 4. 12. *Send demo commands to switch signal relays*

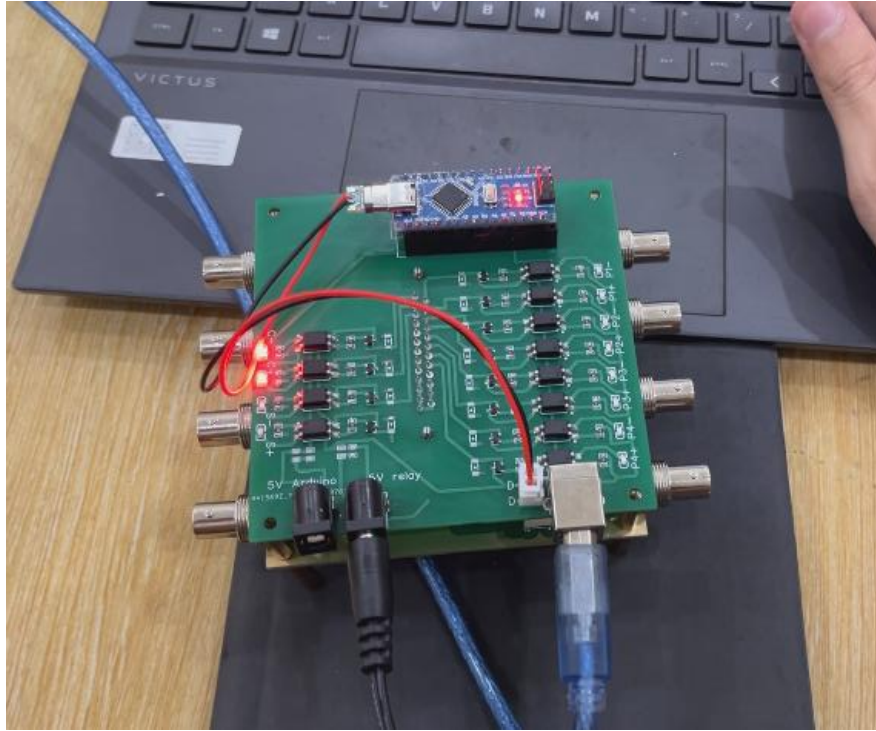


Fig 4. 13. *Two capacitor relays turn on after commands*

Continue to verify other commands, then record the results. For more details, please watch the demo video attached to this report.

After the logic operation is properly reliable, we move to the next phase. This is where we measure parasitic effects and signal integrity. Second phase is the system integration test, or called “cleanroom demo”. Now the test occurs in the highly controlled environment. The below table summarizes items needed for this test phase:

Table 4. 1. List of items needed in the second demo phase

No.	Items	Description
1	Switch box	Switch box and related things such as adapter, USB cable, small wires, ...
2	Test Fixture	Specific test fixture for diode test.
3	Test Instrument	HP 4280A 1MHz C Meter / C-V Plotter Internal DC source: -50VDC Warm-up time: ≥ 30 minutes

4	Test Leads	1m cable length
5	Cleanroom	ISO class 5 (Standard Test & Assembly) Temperature: 23°C $\pm$ 1°C Humidity: 40% - 50% RH

Before conducting the second demo phase, make sure we should follow step-by-step the full cycle C mode measurement procedures which already explained above in the page 13, part 2.1.2 to get the best data record.

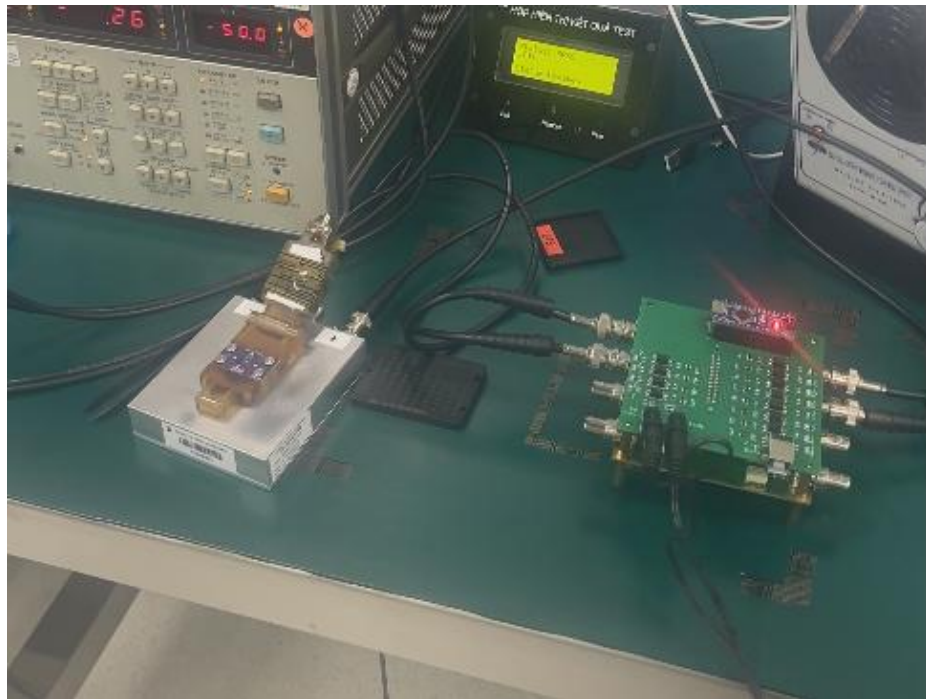


Fig 4. 14. *Setup connection for demo testing*

Use the test cables to connect these devices. To explain more details:

The switch box: connect BNC ports (one positive polarity and one negative polarity at the right of the switch box) to the BNC connectors of the test fixture (one positive polarity and one negative polarity), see the figure above. Ensure that we connect the right polarity to each other carefully. The positive polarity must not be connected to the negative polarity, otherwise it can not work and potentially be damaged. Next, connect the source port (one positive polarity and one negative polarity at the left of the switch box) to the HI and LO terminal of the test instrument HP 4280A. Note that positive port of the switch box is connected to the HI terminal, and the negative one connected to the LO terminal.

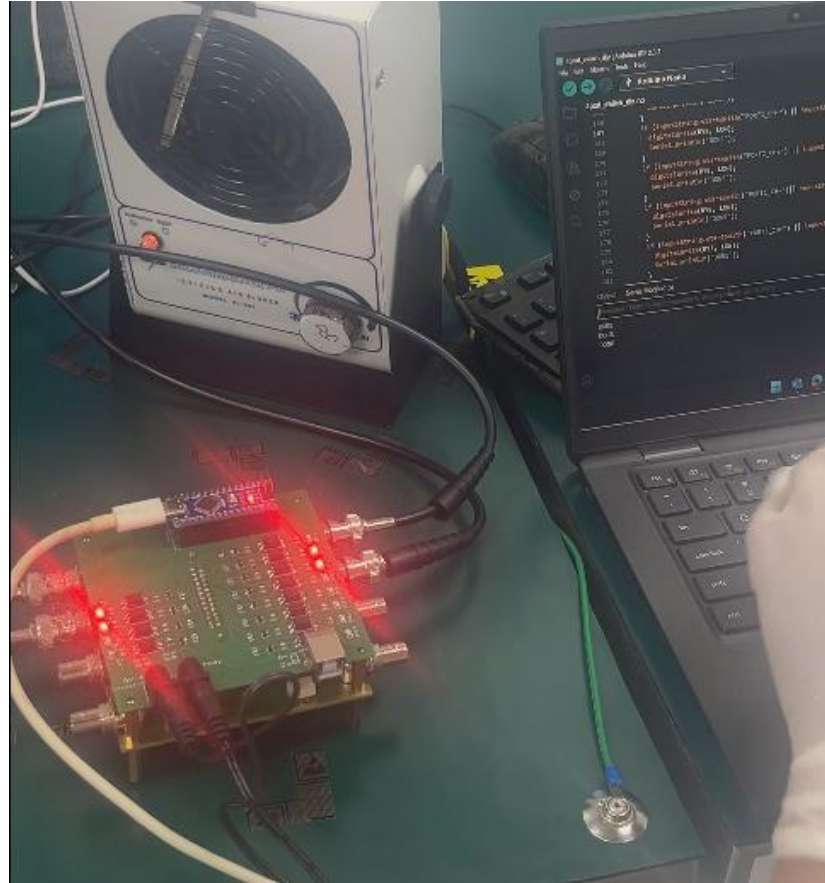


Fig 4. 15. *Send commands to turn on capacitor meter relays and port relays*

After connecting, we control the switch box by Arduino program to switch on the relay for demonstration. In the demo video, we send the input string “CONNECT_CAP” and “PORT1_CH+” and “PORT1_CH-”. Then, press the Reset button of the test instruments HP 4280A and record values in the measurement C mode screen.



Fig 4. 16. Data recorded in capacitance mode on the test instrument's screen

In demo video, we record data 10 times respectively to 10 samples:

$$x = [-0.05, -0.02, -0.06, -0.03, -0.01, -0.04, -0.01, -0.02, -0.04, -0.05] \text{ pF}$$

$$x_1 = -0.05 \text{ pF}$$

$$x_2 = -0.02 \text{ pF}$$

$$x_3 = -0.06 \text{ pF}$$

$$x_4 = -0.03 \text{ pF}$$

$$x_5 = -0.01 \text{ pF}$$

$$x_6 = -0.04 \text{ pF}$$

$$x_7 = -0.01 \text{ pF}$$

$$x_8 = -0.02 \text{ pF}$$

$$x_9 = -0.04 \text{ pF}$$

$$x_{10} = -0.05 \text{ pF}$$

Now, the data is ready to be calculated and analyzed.

4.3. DATA ANALYSIS

After recording 10 datas while testing the test fixture with the switch box in the system, we draw the following line graph to show the results.

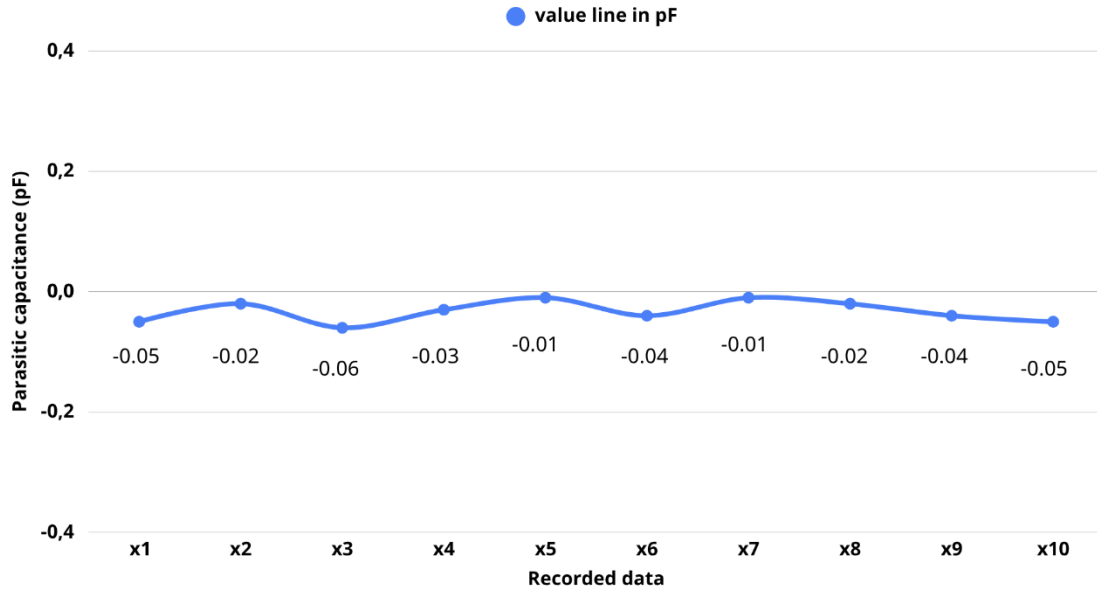


Fig 4. 17. Line graph of 10 datas recorded while testing with the switch box

To prove that the switch box solution helps solve the noise effects in testing, we should apply the Probability and Statistics in data analysis. Statistics makes measurement results more scientific instead of just observing numbers on the 4280A screen. While measuring, the reading result is not a single fixed value. It's affected by many factors because in the pF range very sensitive. Every time measuring, we gain sample from a distribution of possible values. So, to get the central value of repeated measurements, we calculate the Mean. The Mean, or the average value, is calculated by the formula:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{x_1 + x_2 + \dots + x_n}{n}$$

Equation 4. 1. The Mean formula

Here,

$\bar{x}$ is the mean (pF)

x_i is the recorded value at the i time (pF)

n is the total number of observations.

$$\bar{x} = \frac{-(0.05 + 0.02 + 0.06 + 0.03 + 0.01 + 0.04 + 0.01 + 0.02 + 0.04 + 0.05)}{10}$$

$$\Rightarrow \bar{x} = -0.033 \text{ pF}$$

So, the mean is -0.033 pF is the average parasitic noise recorded of the total 10 samples in testing with the switch box. Next, to prove that whether the testing procedure is stable, reliable or not, we must calculate the Sample Standard deviation (s). It is one of the basic methods of statistical analysis. Calculating s allows us to prove that the switch box provides a stably repeatable electrical switching path that does not affect every time switching the relay.

Also, it tells how much the reading values scatter around the mean.

If s is low, small value, $\rightarrow$ the values tend to be close to the mean value. The value is acceptable to be stable and low noise.

If s is high, large value, $\rightarrow$ the values tend to be far from the mean value. The value is unstable and not reliable.

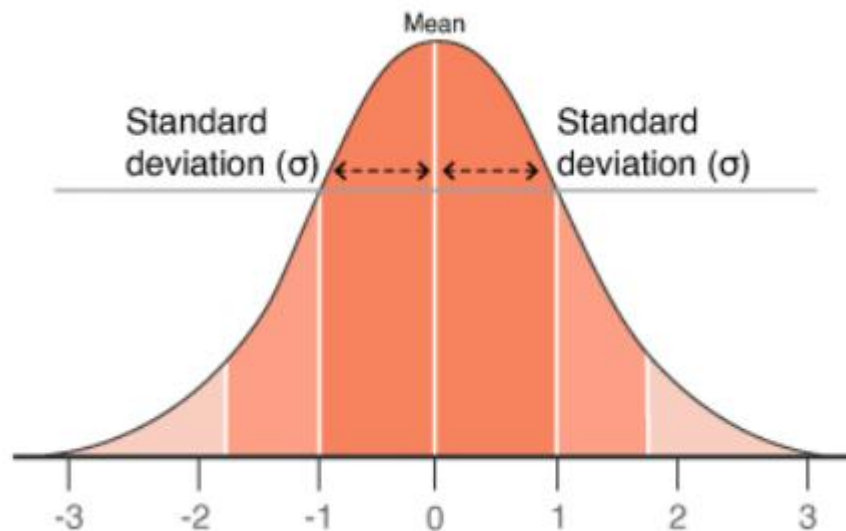


Fig 4. 18. *Standard deviation*

Note that the 10 samples of observation is a small dataset used to estimate the quality of the switch box. It should not be seen as the full population of single cycle measurement. Therefore, the standard deviation is calculated by the formula:

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Equation 4. 2. *Standard Deviation formula*

Here,

s is the sample standard deviation.

$\bar{x}$ is the sample mean (pF)

x_i is the recorded value at the i time (pF)

n is the total number of observations.

Step 1: Calculate the variance (s^2)

$$(-0.05 + 0.033)^2 = 2.89 \times 10^{-4}$$

$$(-0.02 + 0.033)^2 = 1.69 \times 10^{-4}$$

$$(-0.06 + 0.033)^2 = 7.29 \times 10^{-4}$$

$$(-0.03 + 0.033)^2 = 0.09 \times 10^{-4}$$

$$(-0.01 + 0.033)^2 = 5.29 \times 10^{-4}$$

$$(-0.04 + 0.033)^2 = 0.49 \times 10^{-4}$$

$$(-0.01 + 0.033)^2 = 5.29 \times 10^{-4}$$

$$(-0.02 + 0.033)^2 = 1.69 \times 10^{-4}$$

$$(-0.04 + 0.033)^2 = 0.49 \times 10^{-4}$$

$$(-0.05 + 0.033)^2 = 2.89 \times 10^{-4}$$

$$\Rightarrow \text{Sum} = 28.1 \times 10^{-4}$$

$$\Rightarrow s^2 = \frac{28.1 \times 10^{-4}}{10-1} = \frac{0.00281}{9}$$

Step 2: Square root the variance:

$$s = \sqrt{s^2} = \sqrt{\frac{0.00281}{9}} \approx 0.01767 \text{ pF}$$

Based on the specific 10 samples of observation, our sample standard deviation is 0.01767 pF. This value is low enough to be indicated that the switch box is switching signal to be stable and consistent. It solves the problem of fluctuations in reading value in the testing phase.

To make a comparison that the switch box provides the noise-suppressed function, we will test the system without switch box, only the test fixture connected directly to the test instrument through the manually physical plugging/unplugging action. For more details, watch the attached video demo of this project.



Fig 4. 19. Data measurement value while testing with no switch box solution

To prove that without switch box solution, the measurement will be more unreliable and unaccurate due to parasitic noise effects existing the testing, we also calculate and analyze 10 samples of observation in the same procedures above.

In demo video, we record data 10 times respectively to 10 samples:

$$x = [-0.78, 1.56, -1.59, 1.06, -0.84, -1.63, -0.97, -1.54, 1.52, -1.57] \text{ pF}$$

$$x_1 = -0.78 \text{ pF}$$

$$x_2 = 1.56 \text{ pF}$$

$$x_3 = -1.59 \text{ pF}$$

$$x_4 = -1.06 \text{ pF}$$

$$x_5 = -0.84 \text{ pF}$$

$$x_6 = -1.63 \text{ pF}$$

$$x_7 = -0.97 \text{ pF}$$

$$x_8 = -1.54 \text{ pF}$$

$$x_9 = 1.52 \text{ pF}$$

$$x_{10} = -1.57pF$$

We draw the following line graph to show the results.

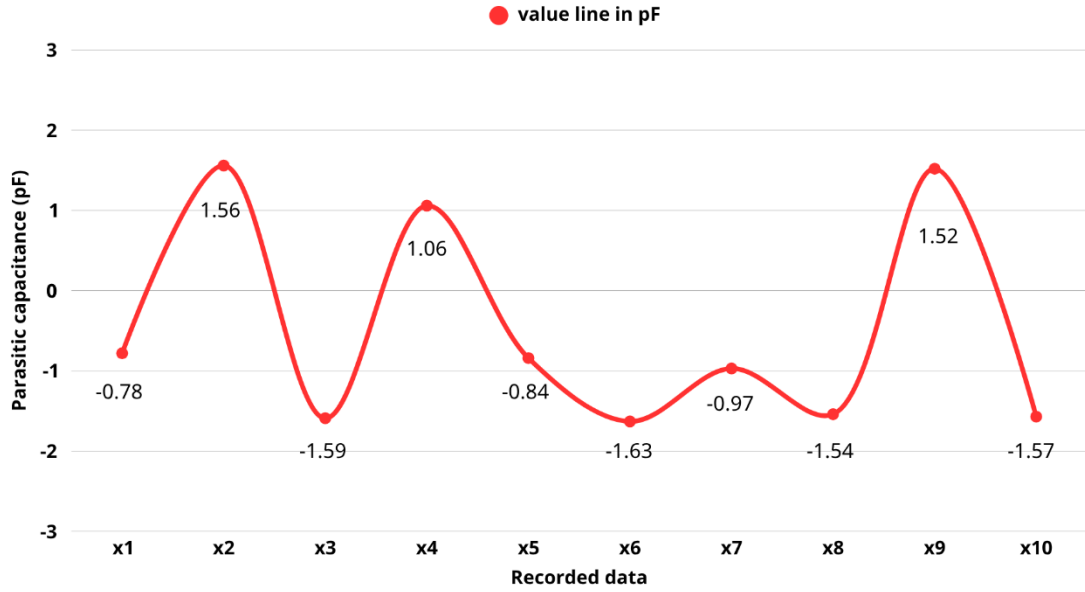


Fig 4. 20. Line graph of 10 datas recorded while testing without the switch box

First, calculating the mean:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{x_1 + x_2 + \dots + x_n}{n}$$

Here,

$\bar{x}$ is the mean (pF)

x_i is the recorded value at the i time (pF)

n is the total number of observations.

$$\bar{x} = \frac{(-0.78 + 1.56 - 1.59 + 1.06 - 0.84 - 1.63 - 0.97 - 1.54 + 1.52 - 1.57)}{10}$$

$$\Rightarrow \bar{x} = -0.478 \text{ pF}$$

Next, calculating the standard deviation:

$$\sigma = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Here,

σ is the standard deviation.

$\bar{x}$ is the sample mean (pF)

x_i is the recorded value at the i time (pF)

n is the total number of observations.

Step 1: Calculate the variance

$$(-0.78 + 0.478)^2 = 0.091204$$

$$(1.56 + 0.478)^2 = 4.153444$$

$$(-1.59 + 0.478)^2 = 1.236544$$

$$(1.06 + 0.478)^2 = 2.365444$$

$$(-0.84 + 0.478)^2 = 0.131044$$

$$(-1.63 + 0.478)^2 = 1.327104$$

$$(-0.97 + 0.478)^2 = 0.242064$$

$$(-1.54 + 0.478)^2 = 1.127844$$

$$(1.52 + 0.478)^2 = 3.992004$$

$$(-1.57 + 0.478)^2 = 1.192464$$

$$\Rightarrow \text{Sum} = 15.85916$$

$$\Rightarrow s^2 = \frac{15.85916}{10-1} = \frac{15.85916}{9}$$

Step 2: Square root the variance:

$$s = \sqrt{s^2} = \sqrt{\frac{15.85916}{9}} \approx 1.327 \text{ pF}$$

Now we make a comparison between the testing with switch box and without switch box. First is the line graph of 10 samples of observation, see the following figure:

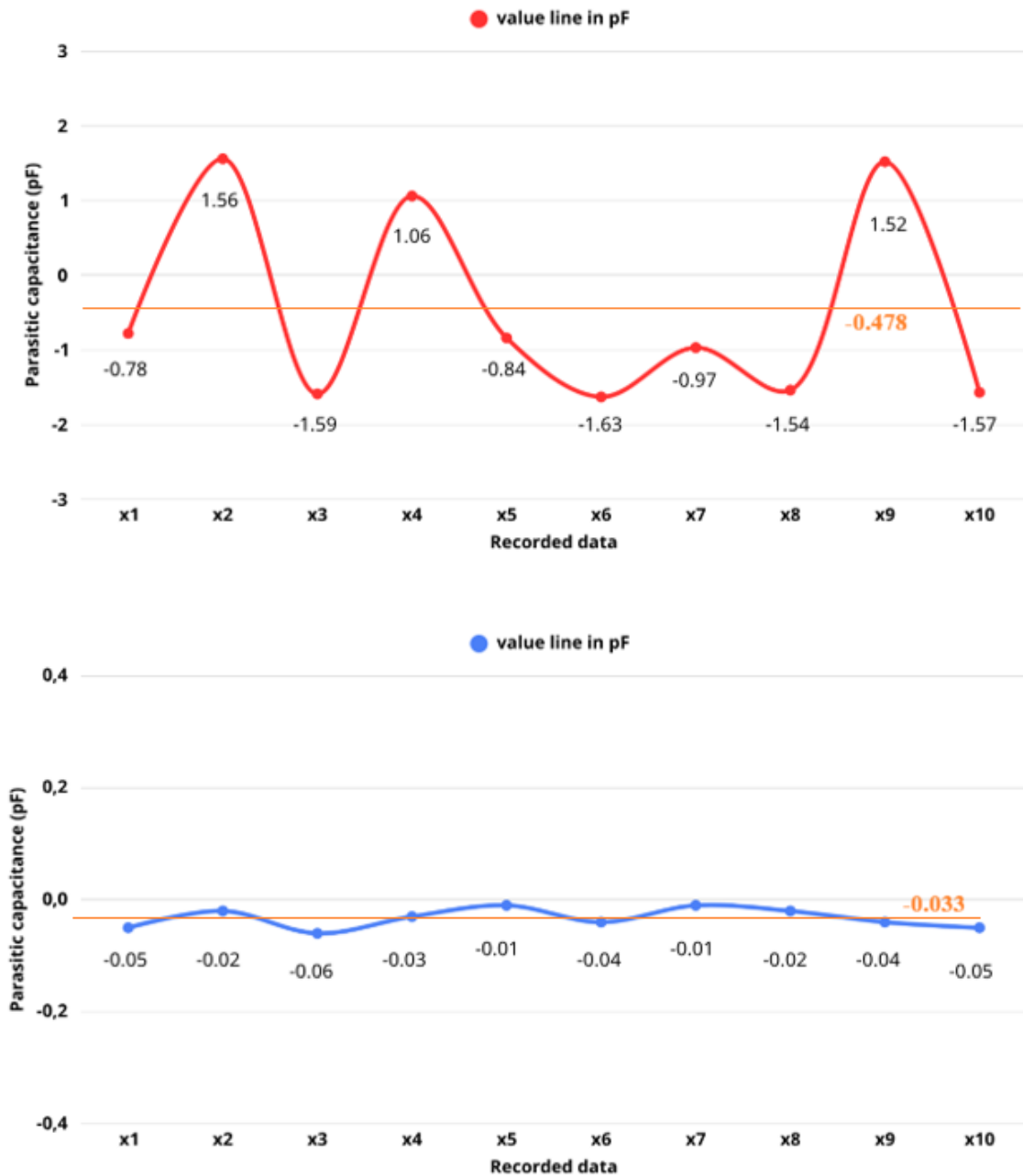


Fig 4. 21. *Comparison between two of line graphs*

Based on 10 samples of measurement data collected while testing in a highly controlled cleanroom, both the line graphs show the negative results. The red line in the top graph jumps massively from -1.63pF to +1.56pF, while measured on a scale of range -3 to +3pF. The standard deviation of this case is around 1.327pF after calculating. This can be called unstable baseline. The main causes for this unstability are human-induced parasitics and existing parasitic effects in the

unshielded cables as well. In the testing production, this result shows the unstable process, which can lead to massive yield loss and high risk of random errors.

Coversely, the blue line graph shows a positive result while testing with the noise-suppressed switch box. The data line is almost flat. All 10 points are tightly closed to 0 between -0.01pF to -0.06pF , while measured on a scale of range -0.4 to $+0.4\text{pF}$. The standard deviation of this case is only 0.01767pF , significantly lower than the other case (1.327pF). This can be called stable high-precision baseline. This stability proves that the hardware architecture successfully isolates the signal path from human-induced interference. By providing a deterministic electrical path, the switch box optimizes the parasitic effects to be as low as possible, gives an more accurate and reliable reading values.

CHAPTER 5. CONCLUSION AND FUTURE DEVELOPMENT

5.1. CONCLUSION

During three months of research and development, this has achieved some fundamental objectives: a deep understanding of the semiconductor testing phase, the existing issue and approaching methods to solve the problem. Also, the project provides a switching architecture solution which empowers the practical hardware implementation together with the optimization in the switch box board design. These studies provide a solid basis for evaluating functional results for measurement and a potential future development. In summary, the project delivers a solid foundation of testing which all intended features of testing can be achieved in the future development.

5.2. PROS AND CONS

The advantages of the project can be explained as follows:

Firstly, the solid fundamental objectives: the studies provide a solid basis to explore the noise-suppressed solution in semiconductor testing. A deep understanding of semiconductor testing procedures, the test instruments and the practical hardware implementation has been achieved.

Secondly, application of the Probability and Statistics in data analysis: to make the research more reliable and scientific, the project has applied the Probability and Statistics while dealing with the several test datas.

Thirdly, optimization in hardware design: the project has successfully applied the hardware optimization in PCB layout from the choice of components, choice of the switching architectures to the strategic layout principles.

Fourthly, optimization in place-suited size. The switch box size is square-shaped box device, suitable for installing and using in the warehouse of manufacturing site.

The disadvantages of the project can be explained as follows:

Firstly, the data analysis is only reliable when the project meets the strict environmental conditions. For example, requirement for temperature and humidity is around at 23 ± 1 °C, 40–60 % RH. If exceeds this specific conditions, the result can be changed. So that reason the test procedures must be performed in the specific cleanroom.

Secondly, the project currently only tests with the one specific type of test fixture used for packaged diode chips.

Thirdly, the switching system has currently not improved the remote controlling function or Internet of Things function. It only controls switching relays by connecting directly to PC through the serial commands.

Fourthly, the switching system currently does not have the back-up function, log history function or real-time clock (RTC) for time tracking.

5.3. FUTURE DEVELOPMENT

To address the current negative aspects of the project, several future developments are proposed:

Firstly, embedded IoT Software Development: The next step will focus on implementing Internet Of Things solution to control the test board through the remote application to execute automated switching procedures, and communicate with a display or data logging system, such as the real-time result output.

Secondly, Data Analysis and Cloud Integration: The testing results will be collected, stored, and visualized via cloud platforms to facilitate remote monitoring and analysis, enhancing the overall utility of the test system.

Thirdly, design optimization: Based on the real-world testing and feedback, hardware improvements will be made to further minimize noise, optimize routing, and possibly reduce PCB size or power consumption.

REFERENCES

- [1]. LetuinEdu, *Hiểu về công đoạn chế tạo bán dẫn*. Hiệu đính: PGS.TS Ngô Ngọc Hà – B.T.T Huyen dịch. Nhà xuất bản Thông tin và truyền thông, 2025.
- [2]. SemiconductorInsight, *Global Semiconductor Packaging and Test Systems Market, Size, Trends, Business Strategies 2025-2032*, SKU: 57409b2aefcc, posted in 2024. URL: [Global Semiconductor Packaging and Test Systems Market 2025](#)
- [3]. IQS directory editorial team, *Cleanrooms: Design, Standards & Applications*, last updated in June 16, 2026. Link: [Cleanrooms: Design, Standards & Applications | IQS Directory](#)
- [4]. Hewlett-Packard company. *Model 4280A 1MHz C Meter/C-V Plotter, Operation and Service Manual*. Manual part no. 04280-90001, microfiche part no. 04280-90051. printed: October 1985. URL: [Model 4280A 1MHz C Meter / C-V Plotter - Operation and Service Manual](#)
- [5]. VSE, *Parasitic Capacitance: High-Speed Circuit Effects*, posted in 2025. URL: [Parasitic Capacitance: High-Speed Circuit Effects - VSE](#)
- [6]. Poulomi Ghosh, *How to handle crosstalk in high-speed PCB designs*. from Sierra Circuits Proto Express, posted in November 1, 2023. URL: [Handling Crosstalk in High-Speed PCB Design | Sierra Circuits](#)
- [7]. Mohamed Faheemuddin, *What is signal propagation delay in PCBs*. from Sierra Circuits Proto Express, posted in October 12, 2023. URL: [What is Signal Propagation Delay in a PCB? | Sierra Circuits](#)
- [8]. Sierra Circuits Proto Express, *How parasitics impact high-speed PCB components*, posted in 2025. URL: [How Parasitics Impact High-Speed Components | Sierra Circuits](#)
- [9]. Changqing Bao, Hongjia Zhu, Jiayu Liu. (2024) “Introduction of parasitic capacitance and methods of reducing its capacitance”. *Proceedings of the 4th International Conference on Materials Chemistry and Environmental*

Engineering. Applied and Computational Engineering, ACE Vol.85, pp.141-147, published in 31 July 2024. URL: <https://doi.org/10.54254/2755-2721/85/20240686>

[10]. National Instruments Corp, *Switching and Multiplexing – Fundamentals of building a test system*. Updated in 2025. URL: [Switching_Multiplexing_Web.pdf](#)

[11]. Metabee Inc, *What is a BNC connector - The Ultimate Guide to Types, Uses & Specification*, posted in June 13, 2025. URL: [What is a BNC Connector? The Ultimate Guide to Types, Uses & Specifications](#)

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[Norlander, Matthew Ryan Keane. "Development and Prototyping of a Ground Fault Circuit Interrupter For 3-Phase 480 Volt Systems.", California Polytechnic State University, 2023](#)

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<https://www.keysight.com/au/en/assets/9018-02732/user-manuals/9018-02732.pdf>

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<https://ijsrem.com/download/railway-track-crack-detection-using-machine-learning/?refresh=697f6158ca16d1769955672&wpdmdl=51039>

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[Vorster, Franzette. "Rekenaargebaseerde instrumentasiestelsel vir halfgeleierkarakterisering", 2014](#)

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[El Sharkawy, A M. M., Sotiriadis, P. P., Bottomley, P. A., Atalar, E.. "Absolute Temperature Monitoring Using RF Radiometry in the MRI Scanner", IEEE, 2006](#)

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<https://www.protoexpress.com/blog/crosstalk-high-speed-pcb-design/>

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<https://thisisbeep.com/what-is-meant-by-propagation-delay/>

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<https://ir.uitm.edu.my/id/eprint/102880/1/102880.pdf>

Trùng khớp < 1% (Internet từ 27-thg 12-2021)

<https://patents.patsnap.com/v/USR46959-enabling-disabling-display-data-channel-access-to-enable-disable-high-bandwidth-digital-content-protection.html>

Trùng khớp < 1% (Ashok Saxena, Ajai Tyagi, D Giri. "Capacitance Transients on Diodes: A Practical Approach to Characterize Levels with Long Time Constants", IETE Journal of Research, 2015)

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<https://b9w.modelcrop.org/sample-standard-deviation-calculation-example>

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<https://era.ed.ac.uk/bitstream/handle/1842/13377/Choi1986.Pdf>

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<https://www.mdpi.com/2304-8158/12/24/4506>

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https://www.ni.com/content/dam/web/scene7/white-papers/22/Fundamentals_of_Building_a_Test_System_Complete_Guide_EN.pdf

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[Jerry C. Whitaker. "Microelectronics", CRC Press, 2018](#)

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[Tokgöz, Furkan. "Analytical Modelling and Multi-Objective Optimization of Axial-Flux Permanent Magnet Machine with Various PCB Stators and Development of a GAN Switched Integrated Motor Drive PCB Motor", Middle East Technical University \(Turkey\), 2024](#)

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<https://trepo.tuni.fi/bitstream/handle/10024/229006/BiruMahlet.pdf?isAllowed=y&sequence=2>

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https://fr.wikipedia.org/wiki/Sigles_de_deux_lettres_suivies_d%27un_chiffre

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https://archive.org/stream/hp_4261A/4261A_djvu.txt

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<http://eprints.umg.ac.id/742/3/BAB%20II.pdf>

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<https://www.ewadirect.com/proceedings/ace/article/view/14516>

Trùng khớp < 1% (P. G. Muzykov. "High field breakdown of narrow quasi uniform field gaps in vacuum", Journal of Applied Physics, 1999)

[P. G. Muzykov. "High field breakdown of narrow quasi uniform field gaps in vacuum", Journal of Applied Physics, 1999](#)

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<https://tailieu.vn/doc/mau-trinh-bay-105391.html>

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http://www.alibaba.com/product-detail/12V-power-switch-button_1730094640.html?s=p

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[F. Cladwell, J.S. Kenney, I.A. Ingram. "Design and implementation of a switched-beam smart antenna for an 802.11b wireless access point", Proceedings RAWCON 2002. 2002 IEEE Radio and Wireless Conference \(Cat. No.02EX573\), 2002](#)

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